Case Studies on

Bivariate Generalized Exponential Distribution under Simple Random and Ranked Set Sampling: Simulation,Estimation, Efficiency Assessment, and Application

Based

on

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by

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Abstract

Bivariate lifetime data frequently contain ties, that is, pairs in which both components fail or occur at exactly the same time, and such data cannot be described by an absolutely continuous bivariate model. The bivariate generalized exponential (BVGE) distribution of Kundu and Gupta has generalized exponential marginals and a singular component along the diagonal, so that it accommodates ties naturally. This report studies the BVGE distribution under two sampling schemes, simple random sampling (SRS) and ranked set sampling (RSS). The joint distribution and density functions are verified numerically, and the marginal and conditional distributions, the hazard functions, the quantile functions, the joint and conditional moments, the covariance and correlation, and the bivariate order statistics with their concomitants are derived and tabulated. Maximum likelihood estimation is developed under SRS and, using the density of the order statistic and its concomitant under the assumption of perfect ranking on the first component, under RSS. Observed information matrices, Wald, log-Wald, profile-likelihood and bootstrap confidence intervals, information criteria and goodness-of-fit tests are reported. The SRS methodology is applied to the UEFA Champions League data ($n = 37$, 14 ties), where the BVGE model is moderately preferred to the Marshall–Olkin model and reproduces the published estimates. Extensive Monte Carlo experiments show that the maximum likelihood estimators are accurate under both schemes and that the log-Wald confidence intervals attain close to the nominal 95% coverage. The RSS estimators are more efficient than their SRS counterparts: the asymptotic determinant-based relative efficiency is 1.12, 1.39 and 1.74 for set sizes 2, 5 and 10, and the finite-sample joint relative efficiency lies between 1.4 and 1.6 for set size 5 and between 1.7 and 2.1 for set size 10, with the largest gains for the rate parameter λ.

Keywords. Bivariate generalized exponential distribution; Singular distribution; Concomitants of order statistics; Maximum likelihood estimation; Ranked set sampling; Relative efficiency; Monte Carlo simulation.

Mathematics Subject Classification. 62H10, 62E15, 62F10, 62F12, 62D05.

1 Introduction

Many applications generate pairs of lifetimes or event times (X_1, X_2) that are dependent and that may take the same value with positive probability: two components of a system that are destroyed by a common shock, the two sensors of a machine that raise an alarm at the same cycle, or the first goal in a football match and the first goal scored by the home side. Absolutely continuous bivariate models assign probability zero to the event $\{X_1 = X_2\}$ and are therefore unsuited to such data. The classical remedy is the Marshall–Olkin bivariate exponential distribution [1], which is obtained from a fatal-shock model and which has a singular component on the diagonal. Its marginals, however, are exponential, with a constant hazard rate, which is restrictive for most lifetime data.

Kundu and Gupta [2] replaced the exponential building blocks by generalized exponential (GE) random variables [3]. If U_1, U_2, U_3 are independent GE variables with common rate λ and shapes $\alpha_1, \alpha_2, \alpha_3$, the pair $X_1 = \max(U_1, U_3)$, $X_2 = \max(U_2, U_3)$ is said to follow the bivariate generalized exponential (BVGE) distribution. The marginals are GE distributions, so that the hazard rates can be increasing, constant or decreasing, the joint distribution function has a simple closed form, and the distribution retains a singular component on the line $x_1 = x_2$, which gives it the Marshall–Olkin tie structure together with flexible marginals. Kundu and Gupta [2] discussed the properties of the model, developed maximum likelihood estimation through an EM algorithm, and fitted the model to the UEFA Champions League data of Meintanis [4]. The correlation between the two components is governed by the same parameter that controls the probability of a tie, a feature that determines for which data the model is appropriate.

Statistics is concerned with drawing inferences about a population from samples. The most basic scheme is simple random sampling (SRS), in which every unit has the same chance of selection. When measuring the study variable is costly or difficult but ranking a small set of units is cheap, McIntyre [8] proposed ranked set sampling (RSS) as an alternative that increases the precision of estimation at the same measurement cost. The mathematical foundations of RSS were developed by Takahasi and Wakimoto [9] and by Dell and Clutter [10], and a comprehensive account is given by Chen et al. [11]. Maximum likelihood estimation under RSS has been studied for many univariate distributions, for example the Pareto distribution [12, 13], the generalized Rayleigh distribution [14], the log-logistic distribution [15], the half-logistic inverted Topp–Leone distribution [16], the exponential Pareto distribution [17], and the power Lindley and weighted Lindley distributions [18]. In all these studies RSS gave more efficient estimators than SRS. In the bivariate setting, ranking on one component and measuring both components of the selected unit yields the order statistic of the ranked component together with its *concomitant* [6]. To the best of our knowledge, a likelihood-based treatment of the BVGE distribution under ranked set sampling has not been reported.

This report follows the structure of the study of the bivariate exponentiated additive Weibull distribution by Poonia et al. [7] and covers the following. First, the joint distribution and density functions of the BVGE distribution are defined and verified numerically. Second, the marginal and conditional distributions, the survival and hazard functions, and the quantile functions are obtained. Third, the joint and conditional moments, the covariance and correlation, and the bivariate order statistics with their concomitants are computed and validated. Fourth, maximum likelihood estimation is developed under SRS and RSS, and the observed Fisher information, confidence intervals, information criteria and goodness-of-fit tests are presented. Fifth, the SRS methodology is applied to the UEFA Champions League data, and a comprehensive Monte Carlo study compares the SRS and RSS estimators through bias, mean squared error, coverage probability and relative efficiency.

The remainder of the report is organized as follows. Section 2 introduces the BVGE distribution and its distributional properties. Section 3 develops estimation under SRS and RSS. Section 4 presents the real data application. Section 5 reports the simulation studies and the efficiency comparison. Section 6 concludes with a summary and directions for further work.

2 The Bivariate Generalized Exponential Distribution

This section defines the BVGE distribution of Kundu and Gupta [2], verifies its distribution and density functions numerically, and derives the marginal and conditional distributions, the survival and hazard functions, the quantile functions, the joint and conditional moments, the covariance and correlation, and the bivariate order statistics with their concomitants. Throughout,

$$A(x) = 1 - e^{-\lambda x}, \qquad s = \alpha_1 + \alpha_2 + \alpha_3,$$

and $\mathrm{GE}(\alpha, \lambda)$ denotes the generalized exponential distribution [3] with distribution function $\left(1 - e^{-\lambda x}\right)^{\alpha}$ and density $\alpha\lambda e^{-\lambda x}\left(1 - e^{-\lambda x}\right)^{\alpha-1}$, $x > 0$.

2.1 Construction, Distribution Function and Density Function

Let $U_1 \sim \mathrm{GE}(\alpha_1, \lambda)$, $U_2 \sim \mathrm{GE}(\alpha_2, \lambda)$ and $U_3 \sim \mathrm{GE}(\alpha_3, \lambda)$ be independent, and define

$$X_1 = \max(U_1, U_3), \qquad X_2 = \max(U_2, U_3). \tag{1}$$

The pair (X_1, X_2) is said to follow the bivariate generalized exponential distribution with parameter vector $\boldsymbol{\theta} = (\alpha_1, \alpha_2, \alpha_3, \lambda)$, written $(X_1, X_2) \sim \mathrm{BVGE}(\alpha_1, \alpha_2, \alpha_3, \lambda)$, with $\alpha_1, \alpha_2, \alpha_3, \lambda > 0$. Since $\{X_1 \le x_1, X_2 \le x_2\} = \{U_1 \le x_1, U_2 \le x_2, U_3 \le \min(x_1, x_2)\}$, the joint cumulative distribution function (CDF) is

$$F(x_1, x_2) = A(x_1)^{\alpha_1}\, A(x_2)^{\alpha_2}\, A(\min(x_1, x_2))^{\alpha_3}, \qquad x_1, x_2 > 0, \tag{2}$$

that is,

$$F(x_1, x_2) = \begin{cases} A(x_1)^{\alpha_1+\alpha_3} A(x_2)^{\alpha_2}, & x_1 < x_2, \\ A(x_1)^{\alpha_1} A(x_2)^{\alpha_2+\alpha_3}, & x_1 > x_2, \\ A(x)^{s}, & x_1 = x_2 = x. \end{cases} \tag{3}$$

The distribution is not absolutely continuous. Its joint probability density function (PDF), taken with respect to Lebesgue measure on $\{x_1 \neq x_2\}$ and with respect to arc length on the diagonal $\{x_1 = x_2\}$, is

$$f(x_1, x_2) = \begin{cases} f_1(x_1, x_2), & x_1 < x_2, \\ f_2(x_1, x_2), & x_1 > x_2, \\ f_0(x), & x_1 = x_2 = x, \end{cases} \tag{4}$$

where

$$f_1(x_1, x_2) = (\alpha_1 + \alpha_3)\, \alpha_2\, \lambda^2\, A(x_1)^{\alpha_1+\alpha_3-1} A(x_2)^{\alpha_2-1} e^{-\lambda(x_1+x_2)}, \tag{5}$$

$$f_2(x_1, x_2) = (\alpha_2 + \alpha_3)\, \alpha_1\, \lambda^2\, A(x_1)^{\alpha_1-1} A(x_2)^{\alpha_2+\alpha_3-1} e^{-\lambda(x_1+x_2)}, \tag{6}$$

$$f_0(x) = \alpha_3\, \lambda\, A(x)^{s-1} e^{-\lambda x}. \tag{7}$$

Integrating the three parts over their supports shows that the probability mass is split as

$$P(X_1 < X_2) = \frac{\alpha_2}{s}, \qquad P(X_1 > X_2) = \frac{\alpha_1}{s}, \qquad P(X_1 = X_2) = \frac{\alpha_3}{s}. \tag{8}$$

The tie probability α_3/s is therefore controlled by α_3 alone, and, as shown in Section 2.6, so is the strength of the dependence between X_1 and X_2.

2.2 Numerical Verification of the Distribution and Density

The expressions (2)–(7) were verified numerically for three illustrative parameter sets, S1, S2 and S3, which cover a low, a high and an intermediate tie probability. The parameter vector fitted to the real data in Section 4 is listed in Table 1 as well and is used in the moment and order-statistic computations of Sections 2.6–2.8. Surfaces of the CDF and of the absolutely continuous part of the PDF are shown in Figures 1 and 2.

Table 1: Parameter sets used throughout the study and the corresponding probability masses in (8). UEFA denotes the maximum likelihood estimate obtained in Section 4.

Set	α_1	α_2	α_3	λ	$P(X_1 < X_2)$	$P(X_1 > X_2)$	$P(X_1 = X_2)$
S1	1.5	1.0	0.5	1.0	0.3333	0.5000	0.1667
S2	1.0	1.0	2.0	1.0	0.2500	0.2500	0.5000
S3	2.0	0.7	0.3	0.5	0.2333	0.6667	0.1000
UEFA	1.4452	0.4681	1.1704	0.03899	0.1518	0.4687	0.3795

The following checks were carried out; their outcomes are summarized in Table 2.

1. The numerical integrals of f_1 over $\{x_1 < x_2\}$, of f_2 over $\{x_1 > x_2\}$ and of f_0 over the diagonal equal α_2/s, α_1/s and α_3/s, respectively, and the total mass equals one.

2. The density is non-negative on its whole support, and the probability of every rectangle $(a, b] \times (c, d]$ computed from the CDF through

$$P(a < X_1 \leq b,\, c < X_2 \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

   is non-negative.

3. The CDF has the correct boundary behaviour, $F(0, x_2) = F(x_1, 0) = 0$ and $F(\infty, \infty) = 1$, and the marginal CDFs obtained as $F(x, \infty)$ and $F(\infty, x)$ agree with the $\mathrm{GE}(\alpha_1 + \alpha_3, \lambda)$ and $\mathrm{GE}(\alpha_2 + \alpha_3, \lambda)$ distributions to a maximum absolute difference of 1.1×10^{-16}.

4. Off the diagonal, the mixed second derivative $\partial^2 F/\partial x_1 \partial x_2$ obtained by finite differences agrees with the absolutely continuous density to a maximum relative error of order 10^{-6}.

5. Samples generated by the exact construction (1) reproduce the theoretical tie probability, and the empirical joint CDF of a large simulated sample differs from F by at most 0.0021 uniformly. Kolmogorov–Smirnov tests do not reject the theoretical marginal distributions.

Table 2: Numerical verification of the BVGE distribution and density functions for the illustrative parameter sets.

Check	S1	S2	S3
Mass of $\{x_1 < x_2\}$ (numerical = theory)	0.3333	0.2500	0.2333
Mass of $\{x_1 > x_2\}$ (numerical = theory)	0.5000	0.2500	0.6667
Mass of $\{x_1 = x_2\}$ (numerical = theory)	0.1667	0.5000	0.1000
Total mass	1.0000	1.0000	1.0000
Minimum of the a.c. density	1.3×10^{-5}	2.9×10^{-6}	6.7×10^{-6}
Minimum rectangle probability	1.2×10^{-8}	9.1×10^{-10}	1.7×10^{-9}
Max. relative error of $\partial^2 F/\partial x_1 \partial x_2$	1.9×10^{-6}	2.6×10^{-6}	1.3×10^{-6}
Simulated $P(X_1 = X_2)$ (theory)	0.1659 (0.1667)	0.5004 (0.5000)	0.1001 (0.1000)
KS p-value, X_1	0.889	0.685	0.096
KS p-value, X_2	0.647	0.536	0.114

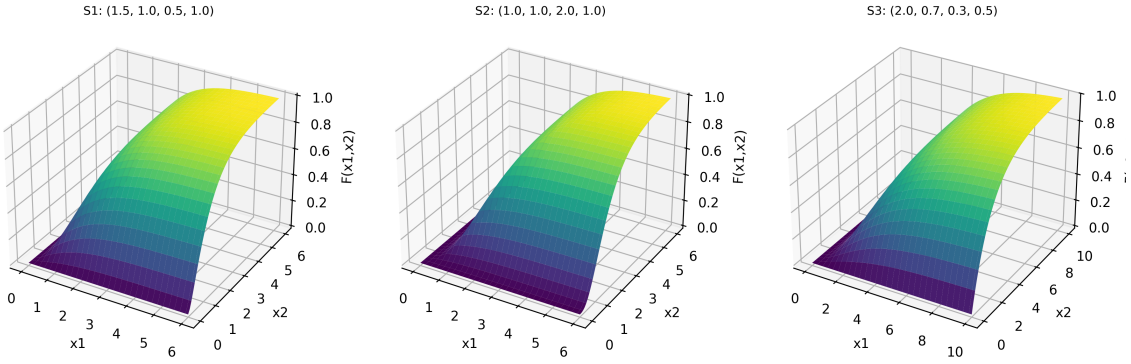


Figure 1: Joint CDF $F(x_1, x_2)$ of the BVGE distribution for the parameter sets S1, S2 and S3 of Table 1.

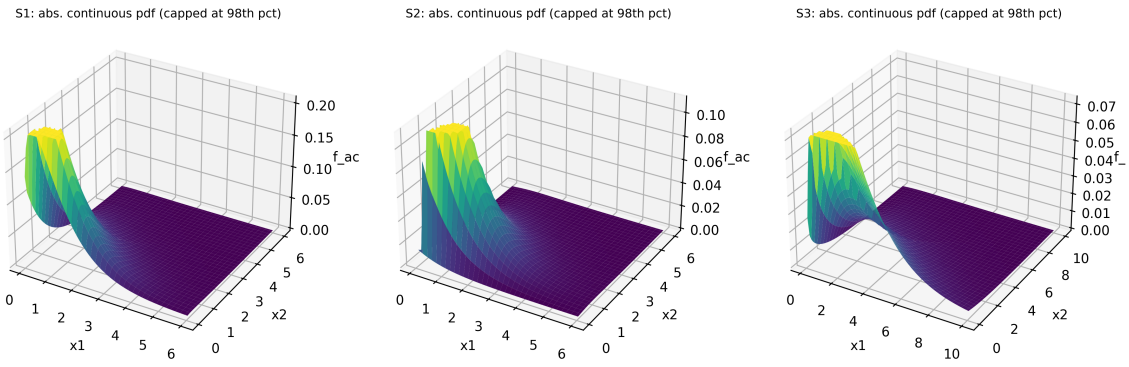


Figure 2: Absolutely continuous part of the joint PDF (off the diagonal) for the parameter sets S1, S2 and S3. The vertical axis is truncated at the 98th percentile of the plotted values.

2.3 Marginal and Conditional Distributions

Marginal distributions. From (3), letting $x_2 \to \infty$ or $x_1 \to \infty$,

$$X_1 \sim \mathrm{GE}(\alpha_1 + \alpha_3, \lambda), \qquad X_2 \sim \mathrm{GE}(\alpha_2 + \alpha_3, \lambda), \tag{9}$$

so that the marginal CDF and PDF of X_i are $F_i(x) = A(x)^{k_i}$ and $f_i(x) = k_i \lambda e^{-\lambda x} A(x)^{k_i - 1}$ with $k_1 = \alpha_1 + \alpha_3$ and $k_2 = \alpha_2 + \alpha_3$. The maximum of the pair has the distribution

$$P(\max(X_1, X_2) \le x) = F(x, x) = A(x)^s, \qquad \max(X_1, X_2) \sim \mathrm{GE}(s, \lambda). \tag{10}$$

Since the marginals are continuous, ranking a sample on X_1 never produces ties, a fact that is used in Section 3 for ranked set sampling.

Conditional distribution. Dividing $\partial F / \partial x_2$ by the marginal density f_2, the conditional distribution function of X_1 given $X_2 = x_2$ is

$$F_{X_1 | X_2}(x_1 \mid x_2) = \begin{cases} \dfrac{\alpha_2}{\alpha_2 + \alpha_3} A(x_1)^{\alpha_1 + \alpha_3} A(x_2)^{-\alpha_3}, & x_1 < x_2, \\ A(x_1)^{\alpha_1}, & x_1 \ge x_2. \end{cases} \tag{11}$$

This distribution is mixed. It has a density

$$f_{X_1 | X_2}(x_1 \mid x_2) = \begin{cases} \dfrac{\alpha_2(\alpha_1 + \alpha_3)}{\alpha_2 + \alpha_3} \lambda e^{-\lambda x_1} A(x_1)^{\alpha_1 + \alpha_3 - 1} A(x_2)^{-\alpha_3}, & x_1 < x_2, \\ \alpha_1 \lambda e^{-\lambda x_1} A(x_1)^{\alpha_1 - 1}, & x_1 > x_2, \end{cases} \tag{12}$$

together with a point mass at $x_1 = x_2$,

$$P(X_1 = X_2 \mid X_2 = x_2) = \frac{\alpha_3}{\alpha_2 + \alpha_3} A(x_2)^{\alpha_1}, \tag{13}$$

which is the jump of (11) at $x_1 = x_2$. The conditional law of X_2 given $X_1 = x_1$ is obtained by interchanging α_1 and α_2 and the roles of x_1 and x_2. The probability of a tie given the value of one component therefore increases with that value, from zero at the origin to $\alpha_3/(\alpha_2 + \alpha_3)$ in the upper tail. Table 3 reports this probability at selected quantiles of X_2.

Table 3: Conditional tie probability $P(X_1 = X_2 \mid X_2 = x_2)$ in (13) at the quantiles of X_2 of order p.

	S1		S2		S3	
p	x_2	atom	x_2	atom	x_2	atom
0.10	0.2426	0.0333	0.6239	0.3094	0.2107	0.0030
0.25	0.5056	0.0833	0.9941	0.4200	0.5754	0.0187
0.50	0.9941	0.1667	1.5784	0.5291	1.3863	0.0750
0.75	1.7457	0.2500	2.3921	0.6057	2.7726	0.1687
0.90	2.6907	0.3000	3.3665	0.6437	4.6052	0.2430

Table 4 reports the numerical verification of (12): the conditional density equals f/f_{X_2} (respectively f/f_{X_1}) off the diagonal up to rounding error, and the density together with the atom (13) integrates to one.

Table 4: Numerical verification of the conditional distributions, hazard functions and quantile functions.

Check	S1	S2	S3
Max. rel. error, $f(x_1 \mid x_2)$ vs f/f_{X_2}	1.4×10^{-15}	1.3×10^{-15}	8.8×10^{-16}
Max. rel. error, $f(x_2 \mid x_1)$ vs f/f_{X_1}	1.2×10^{-15}	1.2×10^{-15}	8.8×10^{-16}
$\max \lvert$total conditional mass $- 1\rvert$	1.1×10^{-16}	1.7×10^{-14}	1.5×10^{-11}
$\max \lvert F(Q(u \mid y) \mid y) - u\rvert$	$0.25^{\dagger}$	1.7×10^{-16}	2.2×10^{-16}
$\max \lvert F(Q(u)) - u\rvert$, marginal	2.2×10^{-16}	2.8×10^{-16}	2.8×10^{-16}
Max. rel. error, h_1 vs $-\partial \ln S/\partial x_1$	3.5×10^{-7}	1.7×10^{-7}	2.4×10^{-7}
Max. rel. error, h_2 vs $-\partial \ln S/\partial x_2$	2.5×10^{-7}	1.7×10^{-7}	2.7×10^{-8}
Minimum of h_1, h_2 (≥ 0)	0.0418	0.0142	0.0540

† The test grid contains the point $u = F_l$ at which the quantile equals y and the CDF jumps by the atom mass (0.25 here); this is a boundary artifact of the check and not an error of the formula.

2.4 Survival and Hazard Functions

The joint survival function $S(x_1, x_2) = P(X_1 > x_1, X_2 > x_2)$ is obtained from the marginal and joint CDFs as

$$S(x_1, x_2) = 1 - A(x_1)^{\alpha_1+\alpha_3} - A(x_2)^{\alpha_2+\alpha_3} + F(x_1, x_2). \tag{14}$$

The marginal survival and hazard functions follow from the GE marginals,

$$S_i(x) = 1 - A(x)^{k_i}, \qquad h_i(x) = \frac{k_i \lambda e^{-\lambda x} A(x)^{k_i-1}}{1 - A(x)^{k_i}}, \qquad i = 1, 2, \tag{15}$$

so that h_i is increasing when $k_i > 1$, constant when $k_i = 1$, decreasing when $k_i < 1$, and tends to λ as $x \to \infty$.

For the bivariate hazard two definitions are used. The Johnson–Kotz hazard gradient [5] is the vector $\mathbf{h}(x_1, x_2) = (h_1^*, h_2^*)$ with $h_i^* = -\partial \ln S/\partial x_i$, that is,

$$h_1^*(x_1, x_2) = \frac{f_1(x_1) - \partial F(x_1, x_2)/\partial x_1}{S(x_1, x_2)}, \qquad h_2^*(x_1, x_2) = \frac{f_2(x_2) - \partial F(x_1, x_2)/\partial x_2}{S(x_1, x_2)}, \tag{16}$$

where, for $x_1 < x_2$,

$$\frac{\partial F}{\partial x_1} = (\alpha_1 + \alpha_3)\lambda e^{-\lambda x_1} A(x_1)^{\alpha_1+\alpha_3-1} A(x_2)^{\alpha_2}, \quad \frac{\partial F}{\partial x_2} = \alpha_2 \lambda e^{-\lambda x_2} A(x_1)^{\alpha_1+\alpha_3} A(x_2)^{\alpha_2-1},$$

and, for $x_1 > x_2$,

$$\frac{\partial F}{\partial x_1} = \alpha_1 \lambda e^{-\lambda x_1} A(x_1)^{\alpha_1 - 1} A(x_2)^{\alpha_2 + \alpha_3}, \quad \frac{\partial F}{\partial x_2} = (\alpha_2 + \alpha_3) \lambda e^{-\lambda x_2} A(x_1)^{\alpha_1} A(x_2)^{\alpha_2 + \alpha_3 - 1}.$$

The scalar bivariate hazard of Basu is, off the diagonal,

$$h(x_1, x_2) = \frac{f(x_1, x_2)}{S(x_1, x_2)}, \qquad x_1 \neq x_2. \tag{17}$$

Table 4 shows that (16) agrees with a finite-difference evaluation of $-\partial \ln S / \partial x_i$ to a relative error of order 10^{-7}, and that both components are non-negative.

Figures 3 and 4 show the bivariate hazard $h(x_1, x_2) = f/S$ for four parameter choices that produce different hazard patterns. Since the marginals are GE distributions, the marginal hazards (15) are always monotone: h_i is decreasing for $k_i < 1$, constant (equal to λ) for $k_i = 1$ and increasing for $k_i > 1$ (Table 5). The two components can therefore have the same or opposite monotonicity, and the surface $h(x_1, x_2)$ reflects this. Unlike lifetime models with a bathtub-shaped hazard, the BVGE distribution does not produce a non-monotone marginal hazard; its flexibility lies in the choice of decreasing, constant or increasing risk for each component and in the tie structure on the diagonal.

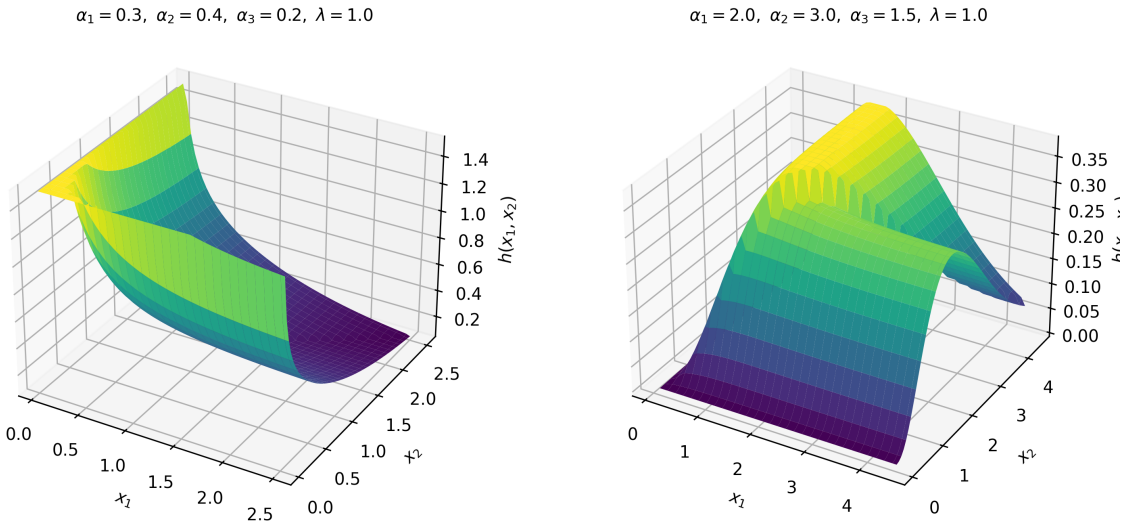


Figure 3: Bivariate hazard rate $h(x_1, x_2)$ of the BVGE distribution. Left: $\alpha_1 = 0.3, \alpha_2 = 0.4, \alpha_3 = 0.2, \lambda = 1$ (both marginal hazards decreasing). Right: $\alpha_1 = 2, \alpha_2 = 3, \alpha_3 = 1.5, \lambda = 1$ (both marginal hazards increasing).

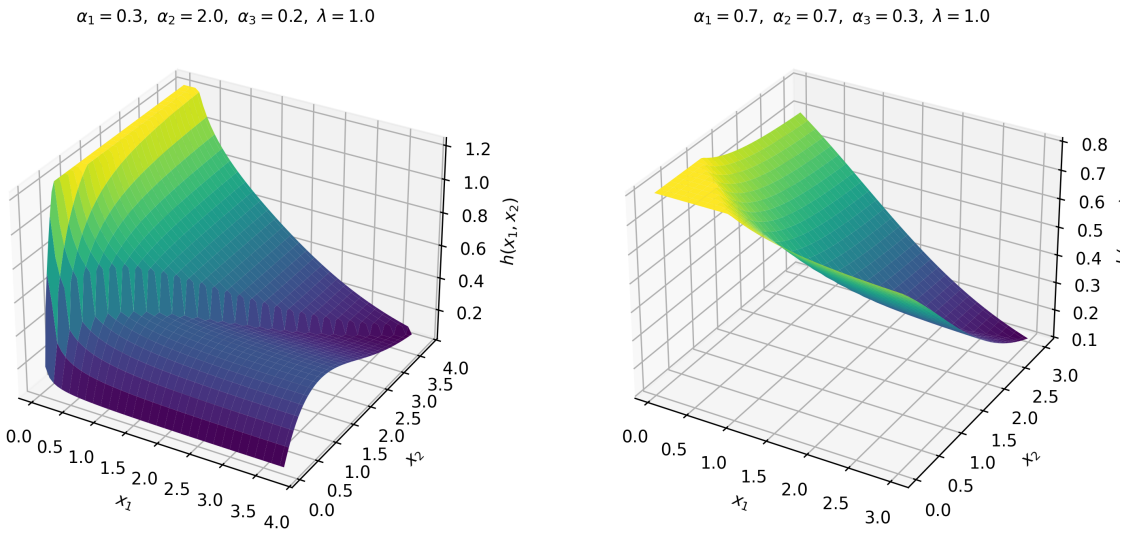


Figure 4: Bivariate hazard rate $h(x_1, x_2)$ of the BVGE distribution. Left: $\alpha_1 = 0.3, \alpha_2 = 2, \alpha_3 = 0.2, \lambda = 1$ (decreasing hazard for X_1, increasing hazard for X_2). Right: $\alpha_1 = 0.7, \alpha_2 = 0.7, \alpha_3 = 0.3, \lambda = 1$ (both marginal hazards constant).

Table 5: Shape of the marginal hazard functions for the parameter choices of Figures 3 and 4 (with $\lambda = 1$ throughout).

Case $(\alpha_1, \alpha_2, \alpha_3)$	Margin	k	Shape	$h(x_{\min})$	$h(x_{\max})$
A-left $(0.3, 0.4, 0.2)$	X_1	0.5	decreasing	5.987	1.022
	X_2	0.6	decreasing	4.286	1.017
A-right $(2.0, 3.0, 1.5)$	X_1	3.5	increasing	9.2×10^{-5}	0.9858
	X_2	4.5	increasing	1.8×10^{-6}	0.9801
B-left $(0.3, 2.0, 0.2)$	X_1	0.5	decreasing	4.974	1.006
	X_2	2.2	increasing	0.0113	0.9860
B-right $(0.7, 0.7, 0.3)$	X_1	1.0	constant	1.000	1.000
	X_2	1.0	constant	1.000	1.000

2.5 Quantile Functions

The marginal CDFs in (9) are invertible in closed form. For $0 < u < 1$,

$$Q_i(u) = -\frac{1}{\lambda} \ln\left(1 - u^{1/k_i}\right), \qquad Q_{\max}(u) = -\frac{1}{\lambda} \ln\left(1 - u^{1/s}\right), \tag{18}$$

for X_i and for $\max(X_1, X_2)$, respectively. Because the conditional distribution (11) has a point mass, its quantile function is piecewise: with $F_l = \frac{\alpha_2}{\alpha_2+\alpha_3} A(y)^{\alpha_1}$ and $F_r = A(y)^{\alpha_1}$,

$$Q_{X_1|X_2}(u \mid y) = \begin{cases} -\dfrac{1}{\lambda} \ln\left(1 - \left[\dfrac{u(\alpha_2+\alpha_3)}{\alpha_2} A(y)^{\alpha_3}\right]^{1/(\alpha_1+\alpha_3)}\right), & 0 < u \le F_l, \\ y, & F_l < u \le F_r, \\ -\dfrac{1}{\lambda} \ln\left(1 - u^{1/\alpha_1}\right), & F_r < u < 1. \end{cases} \tag{19}$$

The conditional quantile function of X_2 given $X_1 = y$ follows by interchanging α_1 and α_2. The marginal quantiles are used for simulation by inversion and for the quantile–quantile plots of

Section 4. Table 6 reports the quantiles of X_1, X_2 and $\max(X_1, X_2)$. The round-trip error $|F(Q(u)) - u|$ is of the order of the machine precision for all sets (Table 4).

Table 6: Quantiles of X_1, X_2 and $\max(X_1, X_2)$ of order p.

Set	Variable	$p = 0.10$	$p = 0.25$	$p = 0.50$	$p = 0.75$	$p = 0.90$
S1	X_1	0.3801	0.6931	1.2279	2.0101	2.9697
	X_2	0.2426	0.5056	0.9941	1.7457	2.6907
	$\max(X_1, X_2)$	0.6239	0.9941	1.5784	2.3921	3.3665
S2	X_1	0.6239	0.9941	1.5784	2.3921	3.3665
	X_2	0.6239	0.9941	1.5784	2.3921	3.3665
	$\max(X_1, X_2)$	0.8263	1.2279	1.8382	2.6679	3.6498
S3	X_1	0.9160	1.5851	2.6926	4.2814	6.2122
	X_2	0.2107	0.5754	1.3863	2.7726	4.6052
	$\max(X_1, X_2)$	1.2478	1.9883	3.1569	4.7842	6.7330

2.6 Joint Moments, Covariance and Correlation

The joint raw moment $E\big[X_1^a X_2^b\big]$ is the sum of the contributions of the three parts of the density (4). With the substitution $u = A(x)$, so that $x = L(u)/\lambda$ with $L(u) = -\ln(1-u)$ and $e^{-\lambda x}dx = du$, the three integrals become integrals over the unit square,

$$E\Big[X_1^a X_2^b\Big] = \lambda^{-(a+b)}\Bigg\{(\alpha_1+\alpha_3)\alpha_2 \int_0^1\!\!\int_0^{u_2} L(u_1)^a L(u_2)^b\, u_1^{\alpha_1+\alpha_3-1} u_2^{\alpha_2-1}\, du_1 du_2 + \alpha_1(\alpha_2+\alpha_3)\int_0^1\!\!\int_0^{u_1} L(u_1)^a L(u_2)^b\, u_1^{\alpha_1-1} u_2^{\alpha_2+\alpha_3-1}\, du_2 du_1 + \alpha_3 \int_0^1 L(u)^{a+b}\, u^{s-1}\, du\Bigg\}. \tag{20}$$

The integrals have no closed form and were evaluated by adaptive numerical quadrature. The marginal moments are available in closed form,

$$E[X_i] = \frac{\psi(k_i+1) - \psi(1)}{\lambda}, \qquad \mathrm{Var}(X_i) = \frac{\psi'(1) - \psi'(k_i+1)}{\lambda^2}, \tag{21}$$

where ψ and ψ' are the digamma and trigamma functions, and they were used to check the numerical results. The covariance and the correlation are

$$\mathrm{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2], \qquad \mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\mathrm{Var}(X_2)}}. \tag{22}$$

Table 7 compares the numerically integrated raw moments with Monte Carlo estimates based on 10^6 simulated pairs. The two agree to within a relative error of 0.4% for the three illustrative sets; for the UEFA set the high-order moments are noisier, as expected for a heavy-tailed quantity, and the largest relative difference is 1.5% for $E[X_1^2 X_2^2]$. Table 8 gives the means, variances, covariance and correlation.

Table 7: Raw joint moments $E[X_1^a X_2^b]$ from numerical integration of (20) (Th.) and from Monte Carlo with 10^6 simulated pairs (MC).

		S1		S2		S3		UEFA	
a	b	Th.	MC	Th.	MC	Th.	MC	Th.	MC
1	0	1.5000	1.4987	1.8333	1.8326	3.2241	3.2262	44.054	44.064
0	1	1.2804	1.2802	1.8333	1.8332	2.0000	2.0014	34.520	34.556
2	0	3.5000	3.4910	4.7222	4.7187	15.560	15.574	2813.2	2819.8
0	2	2.7939	2.7957	4.7222	4.7209	8.0000	8.0111	1971.2	1978.2
1	1	2.2363	2.2347	4.2222	4.2199	7.2481	7.2541	1955.7	1962.1
2	1	6.0272	6.0180	13.389	13.379	39.017	39.051	1.5439×10^5	1.5555×10^5
1	2	5.6889	5.6929	13.389	13.379	33.018	33.073	1.4122×10^5	1.4232×10^5
2	2	18.412	18.431	52.102	52.059	208.74	209.56	1.3921×10^7	1.4127×10^7

Table 8: Means, variances, covariance and correlation of (X_1, X_2), with the Monte Carlo correlation and the tie probability.

Set	$E[X_1]$	$E[X_2]$	$\mathrm{Var}(X_1)$	$\mathrm{Var}(X_2)$	$E[X_1X_2]$	Cov	Corr	Corr (MC)	$P(X_1 = X_2)$
S1	1.5000	1.2804	1.2500	1.1546	2.2363	0.31578	0.26286	0.26275	0.16667
S2	1.8333	1.8333	1.3611	1.3611	4.2222	0.86111	0.63265	0.63170	0.50000
S3	3.2241	2.0000	5.1657	4.0000	7.2481	0.79999	0.17599	0.17403	0.10000
UEFA	44.054	34.520	872.45	779.49	1955.7	434.92	0.52739	0.52700	0.37953

The correlation is not a free parameter: it is driven by α_3, the parameter that also controls the tie probability α_3/s. Table 9 and Figure 5 show that the correlation is close to zero when α_3 is small and tends to one as α_3 grows, and that it tracks the tie probability closely. The BVGE distribution is therefore appropriate for data in which strong dependence is accompanied by a substantial proportion of ties; it cannot represent strongly correlated data with few ties, a feature that guided the choice of the real data set in Section 4. For the UEFA parameters the correlation is 0.527 and the tie probability is 0.380.

Table 9: Correlation and tie probability as functions of α_3 ($\lambda = 1$) for three choices of (α_1, α_2).

	$\mathrm{Corr}(X_1, X_2)$			$P(X_1 = X_2)$		
α_3	$(1, 1)$	$(1.5, 0.5)$	$(0.5, 2)$	$(1, 1)$	$(1.5, 0.5)$	$(0.5, 2)$
0.01	0.0090	0.0101	0.0086	0.0050	0.0050	0.0040
0.05	0.0432	0.0483	0.0411	0.0244	0.0244	0.0196
0.10	0.0826	0.0913	0.0780	0.0476	0.0476	0.0385
0.25	0.1827	0.1969	0.1701	0.1111	0.1111	0.0909
0.50	0.3071	0.3225	0.2831	0.2000	0.2000	0.1667
1	0.4667	0.4788	0.4302	0.3333	0.3333	0.2857
2	0.6327	0.6393	0.5908	0.5000	0.5000	0.4444
4	0.7723	0.7750	0.7358	0.6667	0.6667	0.6154
8	0.8701	0.8710	0.8446	0.8000	0.8000	0.7619
16	0.9300	0.9303	0.9146	0.8889	0.8889	0.8649

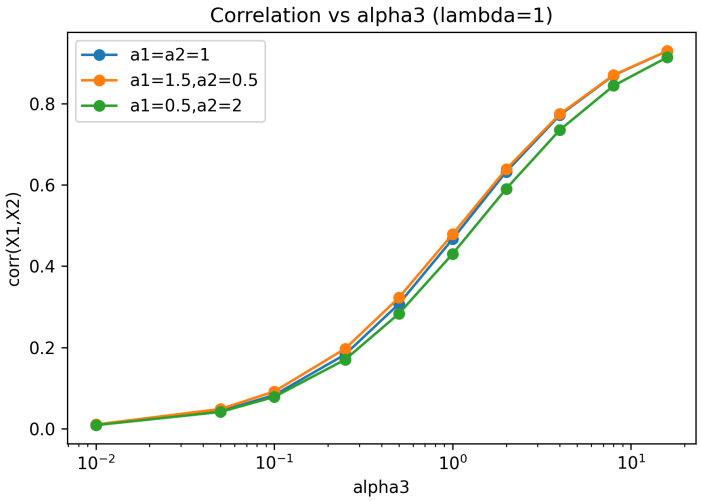


Figure 5: Correlation between X_1 and X_2 as a function of α_3 for three choices of (α_1, α_2) ($\lambda = 1$).

2.7 Conditional Moments

Using the conditional law (11)–(13) and the substitutions $u = A(x_1)$, $v = A(y)$, the conditional moments of X_1 given $X_2 = y$ are

$$E[X_1^a \mid X_2 = y] = \lambda^{-a}\left\{\frac{\alpha_2(\alpha_1+\alpha_3)}{\alpha_2+\alpha_3}\, v^{\alpha_1}\int_0^1 L(vw)^a\, w^{\alpha_1+\alpha_3-1}\, dw + \alpha_1\int_v^1 L(u)^a\, u^{\alpha_1-1}\, du\right\} + \frac{\alpha_3}{\alpha_2+\alpha_3}\, v^{\alpha_1}\, y^a, \tag{23}$$

where the last term is the contribution of the atom at $x_1 = y$. The conditional variance is $\mathrm{Var}(X_1 \mid X_2 = y) = E[X_1^2 \mid X_2 = y] - E[X_1 \mid X_2 = y]^2$, and the moments of X_2 given $X_1 = x$ follow by interchanging α_1 and α_2. The conditional moments must average to the marginal moments, $E\big[E[X_1^a \mid X_2]\big] = E[X_1^a]$ for $a = 1, 2$, and this identity was used as a numerical check (Table 14).

Tables 10–13 report the conditional mean and variance of each component given the other at the quantiles of the conditioning variable, together with the conditional tie probability (13). For all parameter sets the conditional mean increases with the conditioning value, which is the positive dependence induced by the common variable U_3, and it lies above the diagonal for small values of the conditioning variable because of the lower bound created by the atom. The conditional variance is not constant; it is largest at both extremes for the sets with a low tie probability and decreases monotonically as the tie probability becomes dominant (S2).

Table 10: Conditional mean, variance and tie probability, set S1.

	$X_1 \mid X_2 = y$				$X_2 \mid X_1 = x$			
p	y	Mean	Var	Atom	x	Mean	Var	Atom
0.10	0.2426	1.2849	1.1445	0.0333	0.3801	1.0249	0.9607	0.0791
0.25	0.5056	1.3047	1.1077	0.0833	0.6931	1.0735	0.9065	0.1250
0.50	0.9941	1.3831	1.0115	0.1667	1.2279	1.1906	0.8476	0.1768
0.75	1.7457	1.5764	0.9496	0.2500	2.0101	1.3968	0.9303	0.2165
0.90	2.6907	1.8773	1.1974	0.3000	2.9697	1.6613	1.3592	0.2372

Table 11: Conditional mean, variance and tie probability, set S2 (the two conditional laws coincide because $\alpha_1 = \alpha_2$).

p	y	Mean	Var	Atom
0.10	0.6239	1.1308	0.8229	0.3094
0.25	0.9941	1.2950	0.6826	0.4200
0.50	1.5784	1.6244	0.5429	0.5291
0.75	2.3921	2.1521	0.5844	0.6057
0.90	3.3665	2.8164	1.0202	0.6437

Table 12: Conditional mean, variance and tie probability, set S3.

	$X_1 \mid X_2 = y$				$X_2 \mid X_1 = x$			
p	y	Mean	Var	Atom	x	Mean	Var	Atom
0.10	0.2107	3.0003	4.9985	0.0030	0.9160	1.6419	3.2472	0.0647
0.25	0.5754	3.0046	4.9759	0.0188	1.5851	1.7329	3.1344	0.0855
0.50	1.3863	3.0485	4.8001	0.0750	2.6926	1.9100	3.1423	0.1056
0.75	2.7726	3.2471	4.3780	0.1688	4.2814	2.1755	3.6877	0.1195
0.90	4.6052	3.6757	4.5185	0.2430	6.2122	2.4827	5.1943	0.1263

Table 13: Conditional mean, variance and tie probability, UEFA parameters.

	$X_1 \mid X_2 = y$				$X_2 \mid X_1 = x$			
p	y	Mean	Var	Atom	x	Mean	Var	Atom
0.10	7.217	32.466	732.54	0.0937	13.734	19.325	362.82	0.2963
0.25	14.375	33.683	678.73	0.2103	22.777	23.548	322.67	0.3491
0.50	27.295	38.013	549.71	0.3876	37.379	31.023	327.87	0.3953
0.75	46.832	48.167	422.22	0.5542	58.005	41.810	498.91	0.4250
0.90	71.193	63.870	494.81	0.6509	82.882	54.392	969.74	0.4391

Table 14: Total-expectation check: relative error of $E\big[E[X_A^a \mid X_B]\big]$ against $E[X_A^a]$.

Set	Conditioning	$a = 1$	$a = 2$
S1	$X_1 \mid X_2$	6.7×10^{-16}	5.7×10^{-13}
	$X_2 \mid X_1$	3.3×10^{-16}	2.6×10^{-13}
S2	$X_1 \mid X_2$	2.9×10^{-15}	4.1×10^{-13}
	$X_2 \mid X_1$	2.9×10^{-15}	4.1×10^{-13}
S3	$X_1 \mid X_2$	3.6×10^{-15}	2.3×10^{-13}
	$X_2 \mid X_1$	8.9×10^{-16}	3.0×10^{-11}
UEFA	$X_1 \mid X_2$	4.4×10^{-16}	1.6×10^{-12}
	$X_2 \mid X_1$	1.3×10^{-12}	8.0×10^{-12}

2.8 Bivariate Order Statistics and Concomitants

Let (X_{1j}, X_{2j}), $j = 1, \ldots, n$, be a random sample from the BVGE distribution and let $X_{(1:n)} \leq \cdots \leq X_{(n:n)}$ denote the order statistics of the first component. The second component of the unit whose first component has rank r is the *concomitant* $Y_{[r:n]}$ [6]. Since X_1 has a continuous marginal distribution, no ties occur among the first components and the ranking is well defined. The density of $X_{(r:n)}$ is

$$f_{r:n}(x) = C_{r:n}\, F_1(x)^{r-1}\, [1 - F_1(x)]^{n-r}\, f_1(x), \qquad C_{r:n} = \frac{n!}{(r-1)!\,(n-r)!}, \tag{24}$$

with F_1 and f_1 from (9). Equivalently, $T = F_1(X_{(r:n)})$ has a Beta$(r, n-r+1)$ distribution, so that

$$E\Big[X_{(r:n)}^{a}\Big] = \int_0^1 Q_1(t)^a \, \frac{t^{r-1}(1-t)^{n-r}}{B(r, n-r+1)} \, dt, \tag{25}$$

with Q_1 the quantile function in (18). Given $X_{(r:n)} = x$, the concomitant has the conditional distribution of X_2 given $X_1 = x$, which is independent of r and n. Therefore

$$E\Big[Y_{[r:n]}^{b}\Big] = \int_0^\infty E\Big[X_2^b \mid X_1 = x\Big] \, f_{r:n}(x) \, dx, \tag{26}$$

$$E\big[X_{(r:n)} Y_{[r:n]}\big] = \int_0^\infty x \, E[X_2 \mid X_1 = x] \, f_{r:n}(x) \, dx, \tag{27}$$

where the inner conditional moment is (23) with the roles of the components interchanged. The variances, the covariance $\mathrm{Cov}(X_{(r:n)}, Y_{[r:n]}) = E[X_{(r:n)}Y_{[r:n]}] - E[X_{(r:n)}]E[Y_{[r:n]}]$ and the correlation follow. The following identities were used as checks:

$$\sum_{r=1}^{n} E\big[X_{(r:n)}\big] = nE[X_1], \quad \sum_{r=1}^{n} E\Big[Y_{[r:n]}^{b}\Big] = nE[X_2^b], \quad \sum_{r=1}^{n} E\big[X_{(r:n)}Y_{[r:n]}\big] = nE[X_1X_2], \tag{28}$$

the last two holding because the concomitants are a permutation of the second components. Tables 15 and 16 report these quantities for $n = 5$ and $n = 10$ (sets S1 and UEFA), together with Monte Carlo values based on 2×10^5 samples; the tables for S2 and S3 show the same pattern, with correlations between $X_{(r:10)}$ and $Y_{[r:10]}$ increasing from 0.10 to 0.53 for S2 and from 0.03 to 0.13 for S3. The means of $X_{(r:n)}$ and of $Y_{[r:n]}$ both increase with r, the concomitant being less spread out than the order statistic, and the correlation between $X_{(r:n)}$ and $Y_{[r:n]}$ is smallest for the minimum and largest for the maximum. This increasing correlation pattern is what makes ranked set sampling on X_1 informative about the parameters that govern X_2 as well.

Table 15: Order statistics $X_{(r:n)}$ of X_1 and concomitants $Y_{[r:n]}$ of X_2, set S1.

n	r	$E[X_{(r:n)}]$	$\mathrm{Var}[X_{(r:n)}]$	$E[Y_{[r:n]}]$	$\mathrm{Var}[Y_{[r:n]}]$	Cov	Corr	MC $E[Y]$	MC Corr
5	1	0.50635	0.10139	1.0515	0.94383	0.017263	0.055805	1.0475	0.055881
	2	0.88532	0.17538	1.1214	0.90696	0.038412	0.096315	1.1219	0.095905
	3	1.3060	0.29257	1.2182	0.91400	0.073021	0.14121	1.2197	0.14191
	4	1.8734	0.54668	1.3641	1.0430	0.14548	0.19265	1.3646	0.18909
	5	2.9290	1.5498	1.6466	1.7424	0.41302	0.25134	1.6518	0.25200
10	1	0.33377	0.039915	1.0244	0.96589	0.0050819	0.025882	1.0264	0.024525
	2	0.54698	0.057676	1.0537	0.93467	0.0097242	0.041882	1.0558	0.041236
	3	0.74250	0.076209	1.0882	0.90786	0.015138	0.057550	1.0898	0.055624
	4	0.94061	0.098617	1.1290	0.88785	0.021887	0.073967	1.1282	0.075844
	5	1.1533	0.12792	1.1774	0.87860	0.030741	0.091696	1.1771	0.092616
	6	1.3929	0.16916	1.2362	0.88714	0.043053	0.11114	1.2333	0.11211
	7	1.6780	0.23269	1.3100	0.92743	0.061603	0.13261	1.3098	0.13301
	8	2.0428	0.34469	1.4076	1.0317	0.093299	0.15645	1.4064	0.15701
	9	2.5714	0.59547	1.5510	1.2973	0.16153	0.18378	1.5496	0.18489
	10	3.5977	1.5962	1.8261	2.2499	0.42145	0.22239	1.8233	0.22222

Table 16: Order statistics $X_{(r:n)}$ of X_1 and concomitants $Y_{[r:n]}$ of X_2, UEFA parameters.

n	r	$E[X_{(r:n)}]$	$\mathrm{Var}[X_{(r:n)}]$	$E[Y_{[r:n]}]$	$\mathrm{Var}[Y_{[r:n]}]$	Cov	Corr	MC $E[Y]$	MC Corr
5	1	17.128	84.384	21.114	377.46	39.250	0.21992	21.123	0.22046
	2	27.879	131.49	26.258	375.52	66.239	0.29809	26.229	0.30006
	3	39.281	207.62	32.059	437.63	106.93	0.35474	32.049	0.35341
	4	54.279	374.13	39.760	637.74	190.44	0.38988	39.755	0.38775
	5	81.701	1033.4	53.411	1430.8	500.65	0.41174	53.448	0.40983
10	1	12.083	36.905	18.814	384.34	15.662	0.13150	18.769	0.12891
	2	18.461	47.507	21.608	357.37	22.374	0.17171	21.554	0.17167
	3	24.023	59.047	24.269	346.16	29.251	0.20460	24.164	0.20955
	4	29.503	73.358	27.010	348.53	37.378	0.23376	26.960	0.23277
	5	35.274	92.326	29.966	366.68	47.741	0.25947	30.020	0.25768
	6	41.684	119.21	33.289	406.81	61.937	0.28125	33.278	0.28120
	7	49.222	160.81	37.200	482.34	83.194	0.29871	37.210	0.29569
	8	58.778	234.32	42.117	625.04	119.53	0.31233	42.085	0.31112
	9	72.514	399.12	49.042	931.14	198.22	0.32515	49.012	0.32213
	10	98.995	1057.1	61.890	1924.5	501.11	0.35133	62.024	0.35319

Table 17 reports the identity checks (28) for all four sets and the largest deviation from Monte Carlo. The identities hold to numerical precision, and the Monte Carlo means and correlations agree to within sampling error. The componentwise extremes of the pair are also available in closed form. Since $P(\max(X_1, X_2) \le x) = A(x)^s$ by (10), and $P(\min(X_1, X_2) > x) = 1 - A(x)^{k_1} - A(x)^{k_2} + A(x)^s$,

$$E[\max(X_1, X_2)] = \frac{\psi(s+1) - \psi(1)}{\lambda}, \qquad E[\min(X_1, X_2)] = \frac{1}{\lambda} \int_0^1 \frac{1 - u^{k_1} - u^{k_2} + u^s}{1 - u}\, du, \quad (29)$$

and they satisfy $E[\min] + E[\max] = E[X_1] + E[X_2]$ (Table 18).

Table 17: Relative errors of the identities (28) and maximum deviation from Monte Carlo (2×10^5 samples).

Set	n	$\sum E[X_{(r:n)}]$	$\sum E[Y_{[r:n]}]$	$\sum E[Y^2_{[r:n]}]$	$\sum E[XY]$	max rel. MC $E[X]$	max rel. MC $E[Y]$	max abs. MC Corr
S1	5	6.4×10^{-14}	6.0×10^{-13}	2.5×10^{-13}	1.7×10^{-14}	0.0017	0.0039	0.0036
S1	10	3.4×10^{-14}	2.6×10^{-13}	5.9×10^{-13}	5.1×10^{-15}	0.0013	0.0024	0.0019
S2	5	5.1×10^{-14}	3.0×10^{-13}	1.9×10^{-13}	2.3×10^{-13}	0.0013	0.0007	0.0025
S2	10	2.7×10^{-14}	9.1×10^{-14}	8.7×10^{-13}	6.0×10^{-13}	0.0013	0.0038	0.0055
S3	5	5.9×10^{-14}	5.4×10^{-13}	9.7×10^{-13}	1.0×10^{-13}	0.0018	0.0018	0.0012
S3	10	3.6×10^{-14}	1.5×10^{-13}	1.1×10^{-12}	9.8×10^{-14}	0.0007	0.0029	0.0035
UEFA	5	5.7×10^{-14}	5.0×10^{-13}	1.3×10^{-12}	4.5×10^{-13}	0.0012	0.0011	0.0021
UEFA	10	2.7×10^{-14}	9.6×10^{-14}	4.5×10^{-12}	4.0×10^{-13}	0.0024	0.0044	0.0050

Table 18: Expected values of the componentwise maximum and minimum of (X_1, X_2) from (29) and by Monte Carlo (10^6 pairs).

Set	$E[\max]$	MC	$E[\min]$	MC
S1	1.8333	1.8325	0.94704	0.94710
S2	2.0833	2.0832	1.5833	1.5820
S3	3.6667	3.6680	1.5574	1.5585
UEFA	47.617	47.600	30.957	30.941

3 Estimation of the Parameters

This section develops maximum likelihood estimation of $\boldsymbol{\theta} = (\alpha_1, \alpha_2, \alpha_3, \lambda)$ under simple random sampling (SRS) and ranked set sampling (RSS), together with the model selection criteria,

goodness-of-fit procedures, observed information, confidence intervals and the asymptotic measures of efficiency used in Sections 4 and 5.

3.1 Estimation under Simple Random Sampling

Let (x_{1i}, x_{2i}), $i = 1, \ldots, n$, be a simple random sample from the BVGE distribution. Partition the indices into

$$I_0 = \{i : x_{1i} = x_{2i}\}, \quad I_1 = \{i : x_{1i} < x_{2i}\}, \quad I_2 = \{i : x_{1i} > x_{2i}\},$$

with sizes n_0, n_1, n_2, $n_0 + n_1 + n_2 = n$. Using the density (4), the log-likelihood function is

$$\begin{aligned}\ell(\boldsymbol{\theta}) = {} & n_0 \ln(\alpha_3 \lambda) + (s-1) \sum_{i \in I_0} \ln A(x_{1i}) - \lambda \sum_{i \in I_0} x_{1i} \\ & + n_1 \ln\big[(\alpha_1 + \alpha_3)\alpha_2 \lambda^2\big] + (\alpha_1 + \alpha_3 - 1) \sum_{i \in I_1} \ln A(x_{1i}) + (\alpha_2 - 1) \sum_{i \in I_1} \ln A(x_{2i}) - \lambda \sum_{i \in I_1} (x_{1i} + x_{2i}) \\ & + n_2 \ln\big[(\alpha_2 + \alpha_3)\alpha_1 \lambda^2\big] + (\alpha_1 - 1) \sum_{i \in I_2} \ln A(x_{1i}) + (\alpha_2 + \alpha_3 - 1) \sum_{i \in I_2} \ln A(x_{2i}) - \lambda \sum_{i \in I_2} (x_{1i} + x_{2i}).\end{aligned} \tag{30}$$

The likelihood equations $\partial \ell / \partial \boldsymbol{\theta} = \mathbf{0}$ are non-linear and have no closed-form solution, so the maximum likelihood estimate (MLE) $\widehat{\boldsymbol{\theta}}$ is computed numerically. In this study ℓ is maximized with the bound-constrained quasi-Newton algorithm L-BFGS-B [19] on the logarithmic scale $\boldsymbol{\eta} = \ln \boldsymbol{\theta}$, which keeps all parameters positive. Two starting points are used and the better solution is retained: $(1, 1, 1, 1/\bar{x})$, where $\bar{x}$ is the overall mean, and a second point whose shapes are proportional to the observed proportions of I_1, I_2 and I_0, in accordance with (8). A solution is flagged as a boundary solution when a logarithmic parameter reaches its bound.

As an independent check, the EM algorithm of Kundu and Gupta [2, 20] was also implemented. It exploits the construction (1): the maximum of two independent GE variables with common rate λ is again GE, and, given $\max(U_1, U_3) = x$, the event $\{U_1 > U_3\}$ has probability $\alpha_1/(\alpha_1 + \alpha_3)$, irrespective of x. The E-step replaces the unobserved identity of the component that attains each maximum by these conditional probabilities, and the M-step maximizes the resulting weighted generalized exponential likelihood. The EM algorithm and the direct maximization of (30) are compared in Section 4.

When no tie is observed ($n_0 = 0$), the log-likelihood has no term informing α_3 and the maximizer tends to the boundary $\alpha_3 \to 0$. This can occur in small samples when the tie probability α_3/s is small, and its frequency is reported in the simulation study as the proportion of boundary solutions.

Marshall–Olkin comparison model. The BVGE model is compared with the Marshall–Olkin bivariate exponential (MO) distribution [1], the special case with exponential building blocks, with parameters $(\lambda_1, \lambda_2, \lambda_{12})$. Its density is

$$f_{\mathrm{MO}}(x_1, x_2) = \begin{cases} \lambda_1(\lambda_2 + \lambda_{12}) e^{-\lambda_1 x_1 - (\lambda_2 + \lambda_{12}) x_2}, & x_1 < x_2, \\ \lambda_2(\lambda_1 + \lambda_{12}) e^{-(\lambda_1 + \lambda_{12}) x_1 - \lambda_2 x_2}, & x_1 > x_2, \\ \lambda_{12} e^{-(\lambda_1 + \lambda_2 + \lambda_{12}) x}, & x_1 = x_2 = x, \end{cases} \tag{31}$$

with tie probability $\lambda_{12}/(\lambda_1 + \lambda_2 + \lambda_{12})$. It has three parameters and exponential marginals.

3.2 Estimation under Ranked Set Sampling

Ranked set sampling [8] exploits cheap ranking information. In the bivariate setting the units are ranked on one component, here X_1, without measuring it, and both components are then

measured on the selected units. Let m denote the set size and c the number of cycles, so that the RSS sample size is $n = mc$. The procedure is as follows.

1. Draw m^2 units at random and partition them into m sets of m units each.
2. Rank the units within each set on X_1 by an inexpensive criterion, without measuring the variables.
3. From the i-th set ($i = 1, \dots, m$) select the unit of rank i and measure both components, obtaining the pair (x_{ij}, y_{ij}), where x_{ij} is the i-th order statistic of X_1 and y_{ij} is the corresponding concomitant.
4. Repeat steps 1–3 for c cycles to obtain the balanced ranked set sample $\{(x_{ij}, y_{ij}) : i = 1, \dots, m;\ j = 1, \dots, c\}$ of size $n = mc$.

Under perfect ranking, the pairs are independent across ranks and cycles, and, by Section 2.8 and the conditional structure of the model, the pair of rank i has the joint density

$$g_{i:m}(x, y) = C_{i:m}\, F_1(x)^{i-1}\, [1 - F_1(x)]^{m-i}\, f(x, y), \qquad C_{i:m} = \frac{m!}{(i-1)!\,(m-i)!}, \tag{32}$$

because $f_{i:m}(x)\, f(y \mid x) = C_{i:m} F_1(x)^{i-1}[1 - F_1(x)]^{m-i} f_1(x) f(y \mid x)$ and $f_1(x) f(y \mid x) = f(x, y)$; the argument is valid for the mixed distribution, including the pairs on the diagonal. The RSS log-likelihood is therefore

$$\ell_{\mathrm{RSS}}(\boldsymbol{\theta}) = \sum_{j=1}^{c} \sum_{i=1}^{m} \Big\{ \ln C_{i:m} + (i-1) \ln F_1(x_{ij}) + (m-i) \ln[1 - F_1(x_{ij})] + \ln f(x_{ij}, y_{ij}) \Big\}, \tag{33}$$

where $F_1(x) = A(x)^{\alpha_1 + \alpha_3}$ and $\ln f(x_{ij}, y_{ij})$ is the log-density (4) evaluated on the appropriate one of its three parts. Setting $m = 1$ gives the SRS log-likelihood (30). The first term does not depend on $\boldsymbol{\theta}$, and the second and third terms carry the information contained in the ranking. The maximizer of (33) is obtained numerically by the same L-BFGS-B procedure as under SRS. Ranking is on the first component only, and, because X_1 has a continuous marginal, ties never occur in the ranking.

3.3 Information Criteria and Goodness of Fit

Models are compared through

$$\mathrm{AIC} = -2\ell + 2k, \qquad \mathrm{BIC} = -2\ell + k \ln n, \qquad \mathrm{HQIC} = -2\ell + 2k \ln \ln n, \tag{34}$$

[21, 22, 23], where ℓ is the maximized log-likelihood, k the number of parameters (4 for BVGE and 3 for MO) and n the number of pairs. Smaller values indicate a better trade-off between fit and complexity.

Goodness of fit under SRS. If $(X_1, X_2) \sim \mathrm{BVGE}(\boldsymbol{\theta})$ then, by (9) and (10), X_1, X_2 and $\max(X_1, X_2)$ follow $\mathrm{GE}(\alpha_1 + \alpha_3, \lambda)$, $\mathrm{GE}(\alpha_2 + \alpha_3, \lambda)$ and $\mathrm{GE}(s, \lambda)$. The Kolmogorov–Smirnov (KS) statistic [24] $D = \sup_x |F_n(x) - F(x; \widehat{\boldsymbol{\theta}})|$ is computed for each of the three, and p-values are obtained both from the asymptotic KS distribution, which is conservative when the parameters are estimated, and by a parametric bootstrap: B samples are generated from the fitted model, the model is re-fitted to each and the statistic recomputed, giving $p = (1 + \#\{D^* \geq D\})/(B+1)$.

Goodness of fit under RSS. For a ranked set sample the marginals of the pairs are not identically distributed, and a probability-integral-transform (PIT) approach [25] is used instead. If the model is correct and ranking is perfect, then

$$U_{1,ij} = I_{F_1(x_{ij})}(i, m-i+1), \qquad U_{2,ij} = F_{X_2|X_1}(y_{ij} \mid x_{ij}) \tag{35}$$

are independent $U(0,1)$ variables, where $I_z(a,b)$ is the regularized incomplete beta function and, by (11) with the roles of the components interchanged,

$$F_{X_2|X_1}(y \mid x) = \begin{cases} \dfrac{\alpha_1}{\alpha_1+\alpha_3} A(y)^{\alpha_2+\alpha_3} A(x)^{-\alpha_3}, & y < x, \\ A(y)^{\alpha_2}, & y > x, \end{cases}$$

with the tied case $y = x$ randomized, $U_2 = F_l + V(F_r - F_l)$, $V \sim U(0,1)$, $F_l = \frac{\alpha_1}{\alpha_1+\alpha_3} A(x)^{\alpha_2}$ and $F_r = A(x)^{\alpha_2}$. The first transform checks the marginal distribution of the ranked component and the second the dependence structure. KS tests are applied to both, with asymptotic and parametric bootstrap p-values. The same transforms with $m = 1$ can be used under SRS.

3.4 Observed Information and Confidence Intervals

Under standard regularity conditions the MLE is consistent and asymptotically normal,

$$\sqrt{n}\,(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N_4\big(\mathbf{0},\, \mathbf{I}(\boldsymbol{\theta})^{-1}\big)\,, \tag{36}$$

where $\mathbf{I}(\boldsymbol{\theta})$ is the Fisher information per observation [26]. In practice the covariance matrix of $\widehat{\boldsymbol{\theta}}$ is estimated by the inverse of the observed information matrix

$$\mathbf{J}(\widehat{\boldsymbol{\theta}}) = -\left.\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^\top}\right|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}}, \qquad \widehat{\mathrm{Var}}(\widehat{\boldsymbol{\theta}}) = \mathbf{J}(\widehat{\boldsymbol{\theta}})^{-1}, \tag{37}$$

with $\ell = \ell_{\mathrm{SRS}}$ or ℓ_{RSS}. Because of the three-part density and, under RSS, of the rank terms, the Hessian is evaluated numerically by central finite differences with step $10^{-3}|\theta_j|$ (see [16] for the same practice under RSS), and positive definiteness is checked through its eigenvalues. Let $\mathrm{SE}(\widehat{\theta}_j)$ be the square root of the j-th diagonal element of $\mathbf{J}^{-1}$ and z the 97.5th percentile of the standard normal distribution. Four types of 95% confidence interval are used.

1. **Wald:** $\widehat{\theta}_j \pm z\,\mathrm{SE}(\widehat{\theta}_j)$.
2. **Log-Wald** (delta method on the logarithmic scale), which cannot include negative values: $\widehat{\theta}_j \exp\Big\{\pm z\,\mathrm{SE}(\widehat{\theta}_j)/\widehat{\theta}_j\Big\}$.
3. **Profile likelihood:** the set of θ_j with $2\left[\ell(\widehat{\boldsymbol{\theta}}) - \ell_p(\theta_j)\right] \le \chi^2_{1,0.95} = 3.841$ [27], where $\ell_p(\theta_j)$ is the log-likelihood maximized over the remaining parameters.
4. **Nonparametric bootstrap (percentile):** pairs are resampled with replacement B times, the model is re-fitted, and the 2.5th and 97.5th percentiles of the replicates are taken [28].

The Wald interval is symmetric and may be poor for the shape parameters, which have skewed sampling distributions; the log-Wald and profile intervals respect the positivity constraint and are generally preferable.

3.5 Asymptotic Efficiency and Performance Measures

Let $\mathbf{I}_{\mathrm{SRS}}(\boldsymbol{\theta})$ be the Fisher information per observation under SRS, and let

$$\mathbf{I}_{\mathrm{RSS}}(\boldsymbol{\theta}) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{I}_{i:m}(\boldsymbol{\theta}), \qquad \mathbf{I}_{i:m}(\boldsymbol{\theta}) = E\left[-\frac{\partial^2 \ln g_{i:m}(X, Y; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^{\top}}\right] \tag{38}$$

the information per observation under balanced RSS with set size m. For perfect ranking, the RSS information is at least that of SRS in the sense of positive semi-definite ordering [11]. The asymptotic relative efficiencies of RSS with respect to SRS for the same sample size n are

$$\mathrm{ARE}_j = \frac{\left[\mathbf{I}_{\mathrm{SRS}}^{-1}\right]_{jj}}{\left[\mathbf{I}_{\mathrm{RSS}}^{-1}\right]_{jj}}, \quad j = 1, \ldots, 4, \qquad \mathrm{ARE}_{\det} = \left(\frac{\det \mathbf{I}_{\mathrm{RSS}}}{\det \mathbf{I}_{\mathrm{SRS}}}\right)^{1/4}. \tag{39}$$

The information matrices have no closed form and are obtained by evaluating the Hessian of the log-likelihood at the true parameter for a very large simulated sample (10^5 observations or more), so that the Monte Carlo error of $\mathbf{I}$ is below 1%. A value larger than one means that RSS is more efficient.

In the Monte Carlo studies the performance of an estimator $\widehat{\theta}_j$ over N replicates is summarized by

$$\mathrm{Bias} = \frac{1}{N}\sum_{k=1}^{N}\left(\widehat{\theta}_j^{(k)} - \theta_j\right), \qquad \mathrm{MSE} = \frac{1}{N}\sum_{k=1}^{N}\left(\widehat{\theta}_j^{(k)} - \theta_j\right)^2, \tag{40}$$

by the relative bias $100\,\mathrm{Bias}/\theta_j$, by the empirical coverage probability (CP) and average length (AL) of the 95% intervals, by the percentage of boundary solutions, and, to compare the schemes at the same sample size, by the relative efficiency

$$\mathrm{RE}_j = \frac{\mathrm{MSE}_{\mathrm{SRS}}(\widehat{\theta}_j)}{\mathrm{MSE}_{\mathrm{RSS}}(\widehat{\theta}_j)}, \qquad \mathrm{RE}_{\mathrm{joint}} = \left(\frac{\det \widehat{\mathrm{Cov}}_{\mathrm{SRS}}}{\det \widehat{\mathrm{Cov}}_{\mathrm{RSS}}}\right)^{1/4}, \tag{41}$$

where $\widehat{\mathrm{Cov}}$ is the empirical covariance matrix of the four estimates across replicates. Values above one indicate that RSS is more efficient than SRS.

4 Application to the UEFA Champions League Data

4.1 Data

The data set consists of $n = 37$ football matches from the UEFA Champions League of the 2004–2005 and 2005–2006 seasons, in which at least one kick goal and one goal of any type were scored; it was introduced by Meintanis [4] and used by Kundu and Gupta [2] to illustrate the BVGE distribution. The first variable, X_1, is the time (in minutes) of the first goal scored by any team in a kick goal (a penalty or a direct free kick), and the second variable, X_2, is the time of the first goal of any type scored by the home team. A tie, $X_1 = X_2$, occurs when the first goal of any type by the home team is itself the first kick goal of the match. This is the kind of data for which the BVGE distribution is designed: both variables are positive event times, and 14 of the 37 pairs (37.8%) are tied. The data are listed in Table 19. The observed proportions of $\{X_1 < X_2\}$, $\{X_1 > X_2\}$ and $\{X_1 = X_2\}$ are $6/37 = 0.162$, $17/37 = 0.459$ and $14/37 = 0.378$, which are close to the probabilities 0.152, 0.469 and 0.380 implied by the fitted model (Table 1).

Table 19: UEFA Champions League data (minutes): X_1, time of the first kick goal scored by any team; X_2, time of the first goal of any type scored by the home team. Tied pairs are in rows with equal values.

i	X_1	X_2	i	X_1	X_2	i	X_1	X_2	i	X_1	X_2
1	26	20	11	72	72	21	53	39	31	49	49
2	63	18	12	66	62	22	54	7	32	24	24
3	19	19	13	25	9	23	51	28	33	44	30
4	66	85	14	41	3	24	76	64	34	42	3
5	40	40	15	16	75	25	64	15	35	27	47
6	49	49	16	18	18	26	26	48	36	28	28
7	8	8	17	22	14	27	16	16	37	2	2
8	69	71	18	42	42	28	44	13			
9	39	39	19	36	52	29	25	14			
10	82	48	20	34	34	30	55	11			

4.2 Maximum Likelihood Estimation under SRS

The BVGE model was fitted to the data by direct maximization of (30) and by the EM algorithm of Kundu and Gupta [2]. Table 20 shows that the two methods give the same estimates to four decimals, and that these coincide with the published estimates. The maximized log-likelihood is -296.9007 for both algorithms, and it is exactly the same at the published estimates, which confirms the correctness of both the data entry and the likelihood.

Table 20: Maximum likelihood estimates for the UEFA data by direct maximization and by the EM algorithm (40 iterations), compared with Kundu and Gupta [2].

Parameter	Direct ML	EM	Kundu and Gupta
α_1	1.44523	1.44525	1.445
α_2	0.46814	0.46815	0.468
α_3	1.17035	1.17036	1.170
λ	0.03899	0.03899	0.039
$\ell(\widehat{\boldsymbol{\theta}})$	-296.9007	-296.9007	—

Kundu and Gupta report a log-likelihood of -20.59, which differs from -296.90 only because their computation used data rescaled by a factor of 100 (minutes divided by 100). Rescaling the data by a factor c adds the constant $-(2n_1 + 2n_2 + n_0)\ln c$ to the log-likelihood of every model, which equals $-60\ln c$ here. Table 21 shows that the log-likelihood of the fitted model and that at the published estimates both equal -20.5905 for $c = 100$, which agrees with the reported value.

Table 21: Log-likelihood of the BVGE model for the data divided by c, at our estimates and at the estimates of [2].

c	Our estimates	Published estimates
1	-296.9007	-296.9007
10	-158.7456	-158.7456
100	-20.5905	-20.5905

Comparison with the Marshall–Olkin model. The Marshall–Olkin model (31) was fitted to the same data, giving $\widehat{\lambda}_1 = 0.00723$, $\widehat{\lambda}_2 = 0.01635$ and $\widehat{\lambda}_{12} = 0.01768$, with implied tie probability 0.429, against the observed 0.378 and the BVGE value 0.380. The information criteria are given in Table 22. The BVGE model has the smaller AIC, BIC and HQIC; the improvement is 2.33 in AIC, 0.72 in BIC and 1.76 in HQIC, a moderate rather than a decisive preference for the BVGE model. On the scale used by Kundu and Gupta, our Marshall–Olkin log-likelihood is $-299.0672 + 276.3102 = -22.76$, whereas -44.57 is reported there. Because the rescaling constant is the same for both models, the difference in log-likelihood, 2.17, does not depend on the scale, and we have been unable to reproduce the published Marshall–Olkin value; the values of Table 22 are therefore our own.

Table 22: Information criteria for the UEFA data (original scale, $n = 37$).

Model	k	ℓ	AIC	BIC	HQIC
BVGE	4	-296.9007	601.8013	608.2450	604.0730
Marshall–Olkin	3	-299.0672	604.1343	608.9671	605.8381

Goodness of fit. Table 23 gives the KS tests of the three marginal distributions implied by the fitted model (Section 3.3), with asymptotic and parametric bootstrap ($B = 300$) p-values. None of the three is rejected at the 5% level. The bootstrap p-value for $\max(X_1, X_2)$, 0.130, is the smallest of the three and is much lower than its asymptotic value, as expected because the asymptotic KS distribution is conservative when parameters are estimated. Table 24 reports the separate GE fits of the two margins, which agree with the published values $(3.121, 0.0449)$ for X_1 and $(1.678, 0.0413)$ for X_2, and the fitted and empirical distribution functions are shown in Figure 6.

Table 23: Kolmogorov–Smirnov tests of the marginal distributions implied by the fitted BVGE model.

Hypothesis	D	p (asymptotic)	p (parametric bootstrap)
$X_1 \sim \mathrm{GE}(\widehat{\alpha}_1 + \widehat{\alpha}_3, \widehat{\lambda})$	0.1034	0.7861	0.4933
$X_2 \sim \mathrm{GE}(\widehat{\alpha}_2 + \widehat{\alpha}_3, \widehat{\lambda})$	0.1001	0.8172	0.5733
$\max(X_1, X_2) \sim \mathrm{GE}(\widehat{s}, \widehat{\lambda})$	0.1431	0.3969	0.1300

Table 24: Separate generalized exponential fits of the two margins.

Variable	$\widehat{\alpha}$	$\widehat{\lambda}$	ℓ	KS D	p
X_1	3.1193	0.0449	-165.8153	0.1195	0.6228
X_2	1.6775	0.0413	-163.9370	0.1205	0.6131

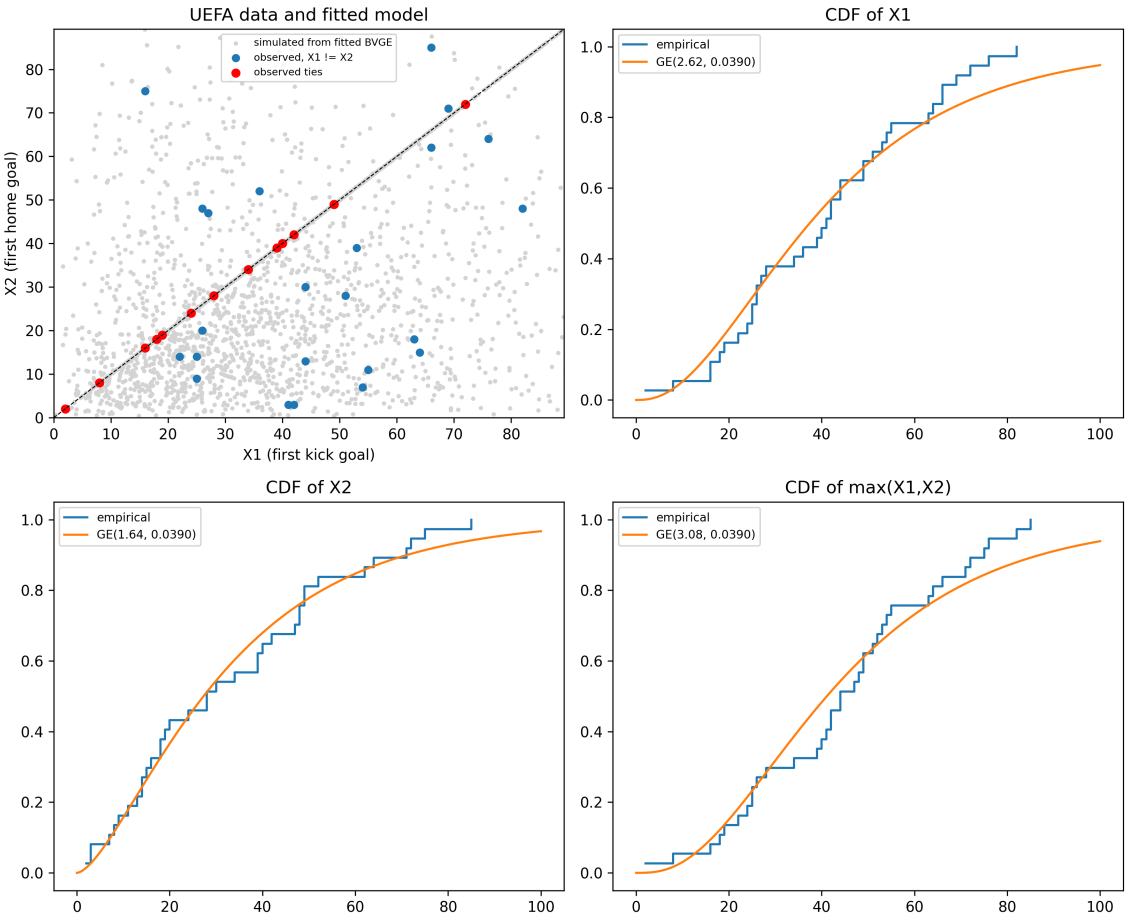


Figure 6: Fitted BVGE model and empirical distribution functions for the UEFA data.

4.3 Observed Information and Confidence Intervals

The observed information matrix at the MLE is given in Table 25. Its eigenvalues are 5.46, 13.35, 30.68 and 69 920.7, all positive, so the matrix is positive definite and the MLE is a proper local maximum. The wide range of eigenvalues reflects the difference in scale between λ and the shape parameters. The correlation matrix of the estimators, obtained from the inverse of the observed information, is given in Table 26; the estimator of λ is positively correlated with all three shape estimators (correlations 0.60, 0.31 and 0.50), whereas $\widehat{\alpha}_2$ is almost uncorrelated with $\widehat{\alpha}_3$.

Table 25: Observed information matrix $\mathbf{J}(\widehat{\boldsymbol{\theta}})$ for the UEFA data.

	α_1	α_2	α_3	λ
α_1	9.016	−0.000	0.877	−421.574
α_2	−0.000	33.711	6.332	−516.447
α_3	0.877	6.332	17.430	−548.611
λ	−421.574	−516.447	−548.611	69910.077

Table 26: Correlation matrix of the estimators for the UEFA data.

	α_1	α_2	α_3	λ
α_1	1.000	0.202	0.246	0.596
α_2	0.202	1.000	-0.060	0.314
α_3	0.246	-0.060	1.000	0.497
λ	0.596	0.314	0.497	1.000

Table 27 reports the standard errors and the four types of 95% interval of Section 3.4, based on $B = 500$ bootstrap samples, together with the intervals published by Kundu and Gupta. The Wald intervals agree closely with the published ones; for example, α_1 has Wald interval $(0.630, 2.261)$ against $(0.657, 2.233)$, and λ has $(0.028, 0.050)$ in both. The log-Wald and profile intervals are shifted to the right and are asymmetric, which is appropriate for the skewed sampling distribution of the shape parameters, and they agree closely with each other. The bootstrap intervals are the widest, in particular for α_1 (0.691 to 3.727) and its bootstrap standard error (0.783) is almost twice the asymptotic one (0.416), which reflects the small sample size and the instability of $\widehat{\alpha}_1$ when a resample contains few observations at large times. The profile deviance curves are shown in Figure 7.

Table 27: Standard errors and 95% confidence intervals for the UEFA data. The last column gives the intervals published in [2].

Parameter	MLE	SE	Boot SE	Wald	Log-Wald	Profile	Bootstrap	Published
α_1	1.4452	0.4160	0.7833	(0.630, 2.261)	(0.822, 2.541)	(0.800, 2.469)	(0.691, 3.727)	(0.657, 2.233)
α_2	0.4681	0.1880	0.2351	(0.100, 0.837)	(0.213, 1.029)	(0.189, 0.938)	(0.162, 1.101)	(0.167, 0.769)
α_3	1.1704	0.2867	0.3125	(0.609, 1.732)	(0.724, 1.892)	(0.698, 1.832)	(0.771, 1.940)	(0.651, 1.689)
λ	0.0390	0.0056	0.0057	(0.028, 0.050)	(0.029, 0.052)	(0.029, 0.051)	(0.032, 0.053)	(0.028, 0.050)

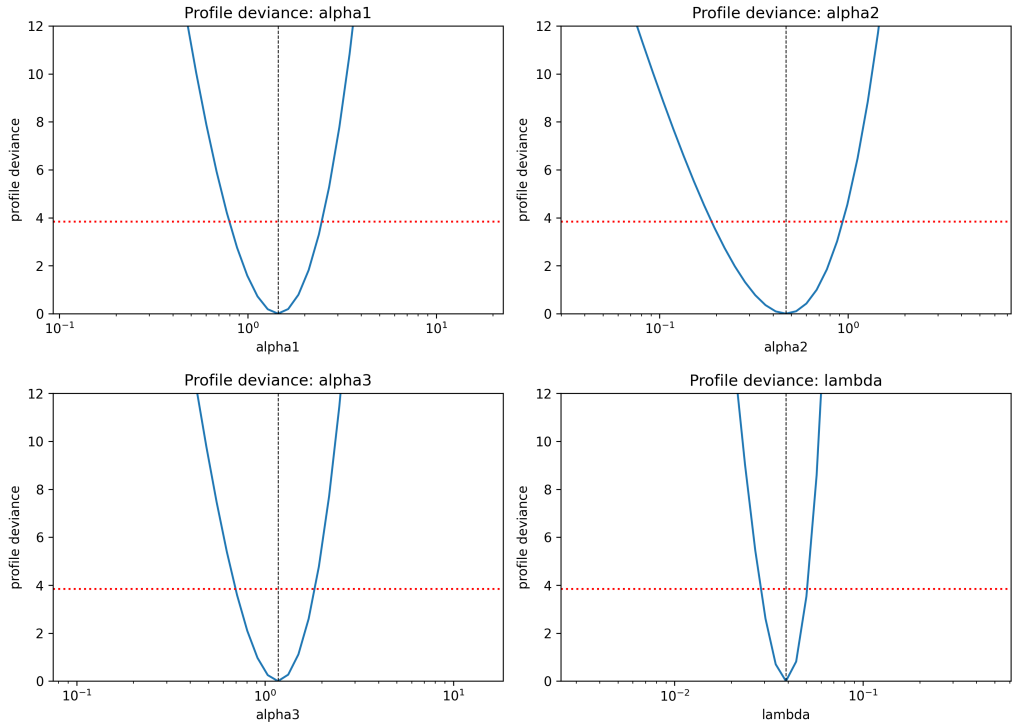


Figure 7: Profile deviance of each parameter for the UEFA data; the horizontal line is $\chi^2_{1,0.95} = 3.841$ and its intersections with the curve give the profile-likelihood intervals.

Ranked set sampling. The real data cannot be analysed under RSS: they form a simple random sample of 37 pairs and no ranking information was collected, and forming ranked sets from such a small sample would waste most of the observations. The ranked-set methodology is therefore illustrated and evaluated on samples simulated from the model fitted in this section, in Section 5.

5 Simulation Studies

This section reports the simulation studies. All samples are generated exactly from the construction (1). The parameter sets are S1–S3 and the UEFA estimate of Section 4 (Table 1); the UEFA set is the main one, since it is estimated from real data. Section 5.1 studies the SRS estimators, Section 5.2 illustrates estimation and inference under RSS on one simulated sample, Section 5.3 gives the asymptotic efficiency, and Section 5.4 compares RSS and SRS through Monte Carlo experiments.

5.1 Monte Carlo Study under SRS

For each parameter set and each $n \in \{30, 50, 70, 100\}$, $R = 1000$ SRS samples were generated, the model was fitted by maximum likelihood, and Wald and log-Wald intervals were computed from the observed information. Table 28 gives bias, MSE, coverage and average length for the UEFA set, and Table 29 gives the relative bias and log-Wald coverage for S1–S3 at $n = 30$ and $n = 100$.

The estimators are positively biased in small samples (relative bias 5–14% at $n = 30$ for the shape parameters), and the bias and MSE fall steadily with n: at $n = 100$ the relative bias is 1–3% and the MSE is roughly proportional to $1/n$. Coverage is close to 0.95 for both interval types from $n = 50$ onward (Wald 0.91–0.96, log-Wald 0.93–0.96 over all cells), and the interval lengths shrink with n (Figure 8). The mean number of ties grows with n and with the tie probability: for the UEFA set it is 11.3, 19.1, 26.6 and 37.8 for $n = 30, 50, 70, 100$, against 3.0 to 10.0 for S3. The proportion of boundary solutions, which are excluded from the bias and MSE summaries, is at most 0.3% except for S3 at small n (4.0%, 0.6% and 0.2% for $n = 30, 50, 70$). This agrees with the probability $0.9^{30} = 0.042$ of observing no tie in 30 pairs when the tie probability is 0.1 (Section 3.1).

Table 28: SRS Monte Carlo, UEFA parameters ($R = 1000$): relative bias (%), MSE, Wald and log-Wald coverage and log-Wald average length.

n	Param.	Rel. bias	MSE	CP Wald	CP log-Wald	AL log-Wald
30	α_1	12.68	0.3707	0.953	0.946	2.224
	α_2	13.37	0.0692	0.938	0.944	1.086
	α_3	8.76	0.1577	0.959	0.945	1.463
	λ	5.38	5.40×10^{-5}	0.951	0.935	0.0263
50	α_1	6.02	0.1669	0.945	0.945	1.567
	α_2	4.15	0.0340	0.912	0.956	0.738
	α_3	4.60	0.0773	0.955	0.945	1.050
	λ	2.80	2.55×10^{-5}	0.958	0.952	0.0199
70	α_1	4.63	0.1174	0.942	0.933	1.285
	α_2	3.73	0.0248	0.928	0.945	0.607
	α_3	3.40	0.0498	0.964	0.959	0.869
	λ	2.06	1.96×10^{-5}	0.940	0.935	0.0167
100	α_1	3.20	0.0715	0.948	0.950	1.049
	α_2	2.92	0.0165	0.938	0.937	0.494
	α_3	1.87	0.0355	0.951	0.946	0.712
	λ	1.09	1.26×10^{-5}	0.955	0.945	0.0138

Table 29: SRS Monte Carlo, sets S1–S3: relative bias (%) and log-Wald coverage (CP).

		α_1		α_2		α_3		λ	
Set	n	bias	CP	bias	CP	bias	CP	bias	CP
S1	30	9.43	0.954	9.44	0.935	10.28	0.948	5.40	0.931
S1	100	2.93	0.958	2.01	0.942	1.72	0.959	1.20	0.952
S2	30	14.57	0.941	13.51	0.932	12.11	0.938	5.11	0.919
S2	100	3.25	0.944	3.33	0.942	2.80	0.963	0.85	0.951
S3	30	9.86	0.938	5.92	0.947	10.16	0.953	4.75	0.940
S3	100	2.88	0.948	2.26	0.949	1.61	0.950	1.56	0.944

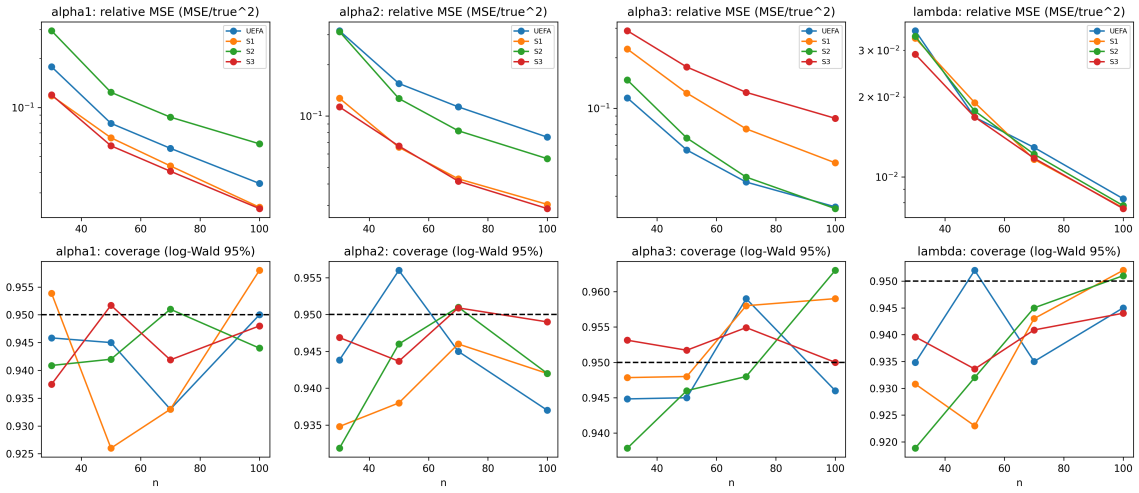


Figure 8: SRS Monte Carlo: relative MSE (top, logarithmic scale) and log-Wald coverage (bottom) against n for the four parameter sets.

5.2 Estimation and Inference under RSS

One RSS sample ($m = 5$, $c = 8$, $n = 40$) and one independent SRS sample of the same size were simulated from the UEFA model, with 14 and 9 ties, respectively. The RSS sample was fitted by maximizing (33). Tables 30–33 give the estimates, standard errors, intervals, information criteria, goodness-of-fit tests (Section 3.3, $B = 200$ bootstrap samples) and the observed information matrix.

In this sample the RSS standard errors are smaller than the SRS ones for α_1 (SE ratio 1.50), λ (1.09) and α_2 (1.01), but not for α_3 (0.81). The SRS sample contained only 9 ties against about 15 expected, which pulled $\widehat{\alpha}_3$ down to 0.62 and its intervals away from the true value 1.170; this is a single-sample effect, and one sample cannot measure efficiency, which is done in Sections 5.3 and 5.4. The BVGE model is preferred to Marshall–Olkin under both designs (AIC 664.8 against 692.0 for RSS and 678.3 against 694.4 for SRS; the RSS and SRS log-likelihoods are not comparable with each other because the RSS one contains the rank terms), and none of the four PIT tests is rejected (bootstrap p-values between 0.41 and 0.81). The RSS observed information is positive definite, with eigenvalues 10.70, 20.12, 36.23 and 137 401.

Table 30: Estimates, standard errors and 95% intervals for an RSS ($m = 5$, $c = 8$) and an SRS sample of size 40 from the UEFA model.

Design	Param.	MLE	SE	Wald	Log-Wald	Profile	True
RSS	α_1	1.2703	0.3004	(0.681, 1.859)	(0.799, 2.019)	(0.776, 1.966)	1.4452
	α_2	0.5250	0.1765	(0.179, 0.871)	(0.272, 1.015)	(0.248, 0.946)	0.4681
	α_3	0.9848	0.2222	(0.549, 1.420)	(0.633, 1.532)	(0.610, 1.484)	1.1704
	λ	0.0423	0.0049	(0.033, 0.052)	(0.034, 0.053)	(0.033, 0.052)	0.0390
SRS	α_1	1.7428	0.4505	(0.860, 2.626)	(1.050, 2.893)	(1.029, 2.834)	1.4452
	α_2	0.6292	0.1780	(0.280, 0.978)	(0.361, 1.096)	(0.339, 1.042)	0.4681
	α_3	0.6216	0.1809	(0.267, 0.976)	(0.351, 1.099)	(0.325, 1.037)	1.1704
	λ	0.0384	0.0053	(0.028, 0.049)	(0.029, 0.050)	(0.029, 0.049)	0.0390

Table 31: Information criteria for the samples of Table 30.

Design	Model	k	ℓ	AIC	BIC	HQIC
RSS	BVGE	4	-328.417	664.834	671.590	667.277
RSS	Marshall–Olkin	3	-342.996	691.991	697.058	693.823
SRS	BVGE	4	-335.160	678.321	685.076	680.763
SRS	Marshall–Olkin	3	-344.216	694.433	699.499	696.265

Table 32: PIT-based KS tests: U_1 (ranked margin) and U_2 ($X_2 \mid X_1$).

Design	D_1	p_1 (asym.)	p_1 (boot.)	D_2	p_2 (asym.)	p_2 (boot.)
RSS	0.0916	0.8601	0.6468	0.0803	0.9405	0.8109
SRS	0.1065	0.7145	0.4080	0.0963	0.8178	0.6318

Table 33: Observed information matrix for the RSS sample.

	α_1	α_2	α_3	λ
α_1	25.624	0.000	14.469	-1371.007
α_2	0.000	36.927	7.897	-547.535
α_3	14.469	7.897	36.803	-1502.626
λ	-1371.007	-547.535	-1502.626	137369.202

5.3 Asymptotic Properties and Efficiency

Table 34 gives the expected information per observation, estimated from 10^5 simulated observations at the UEFA parameters, the resulting asymptotic variances and the AREs (39) for $m = 5$. RSS is asymptotically more efficient for every parameter: ARE = 1.29, 1.07, 1.17 and 1.57 for $\alpha_1, \alpha_2, \alpha_3, \lambda$, and $\mathrm{ARE}_{\mathrm{det}} = 1.39$, so RSS reaches the same overall precision with about 28% fewer observations. The gain is largest for λ and α_1, which are informed by the ranked component, and smallest for α_2, which enters only the second component. At $n = 40$ the asymptotic standard errors are $(0.354, 0.180, 0.256, 0.0044)$ under RSS and $(0.402, 0.186, 0.277, 0.0055)$ under SRS, of the same order as the standard errors of Table 30.

Table 34: Expected information per observation and asymptotic variances, UEFA parameters, SRS and RSS ($m = 5$).

Param.	$I_{\mathrm{RSS},jj}$	$I_{\mathrm{SRS},jj}$	$\mathrm{AVar}_{\mathrm{RSS}}$	$\mathrm{AVar}_{\mathrm{SRS}}$	ARE_j
α_1	0.48244	0.24643	5.01901	6.46231	1.2876
α_2	0.86831	0.86932	1.30032	1.39036	1.0692
α_3	0.71048	0.47372	2.61986	3.06091	1.1684
λ	4556.88	1847.54	0.00078	0.00122	1.5717
Determinant-based ARE					1.3942

Table 35 and Figure 9 extend this to $m = 1, \ldots, 10$ and to all parameter sets. The $m = 1$ row, computed from an independent SRS replicate, lies between 0.99 and 1.01, which shows that the Monte Carlo noise of the method is below 1%. The determinant-based gain increases steadily with m (1.12, 1.39 and 1.74 for $m = 2, 5, 10$ for the UEFA set) and is nearly the same for all four parameter sets. At $m = 10$ the per-parameter AREs for UEFA, S1, S2 and S3 are 1.56, 1.70, 1.33, 1.88 for α_1; 1.09, 1.15, 1.17, 1.10 for α_2; 1.30, 1.13, 1.56, 1.07 for α_3; and 2.16, 1.94, 2.13, 2.00 for λ.

Table 35: Asymptotic relative efficiency of RSS against SRS by set size m: UEFA parameters (per parameter and determinant-based) and determinant-based ARE for S1–S3.

	UEFA					$\mathrm{ARE}_{\mathrm{det}}$		
m	α_1	α_2	α_3	λ	det	S1	S2	S3
1	1.0023	0.9866	1.0022	0.9999	0.9977	0.9993	1.0019	1.0003
2	1.0828	1.0243	1.0493	1.1621	1.1186	1.1181	1.1252	1.1217
3	1.1545	1.0399	1.0942	1.3055	1.2194	1.2188	1.2268	1.2245
4	1.2252	1.0541	1.1301	1.4390	1.3101	1.3065	1.3201	1.3130
5	1.2927	1.0559	1.1644	1.5675	1.3907	1.3876	1.3981	1.3969
6	1.3465	1.0666	1.1979	1.6888	1.4668	1.4624	1.4769	1.4704
7	1.4059	1.0653	1.2280	1.8122	1.5363	1.5339	1.5511	1.5452
8	1.4571	1.0821	1.2521	1.9291	1.6063	1.5990	1.6148	1.6152
9	1.5096	1.0849	1.2740	2.0434	1.6691	1.6615	1.6866	1.6783
10	1.5588	1.0930	1.2962	2.1584	1.7314	1.7215	1.7443	1.7390

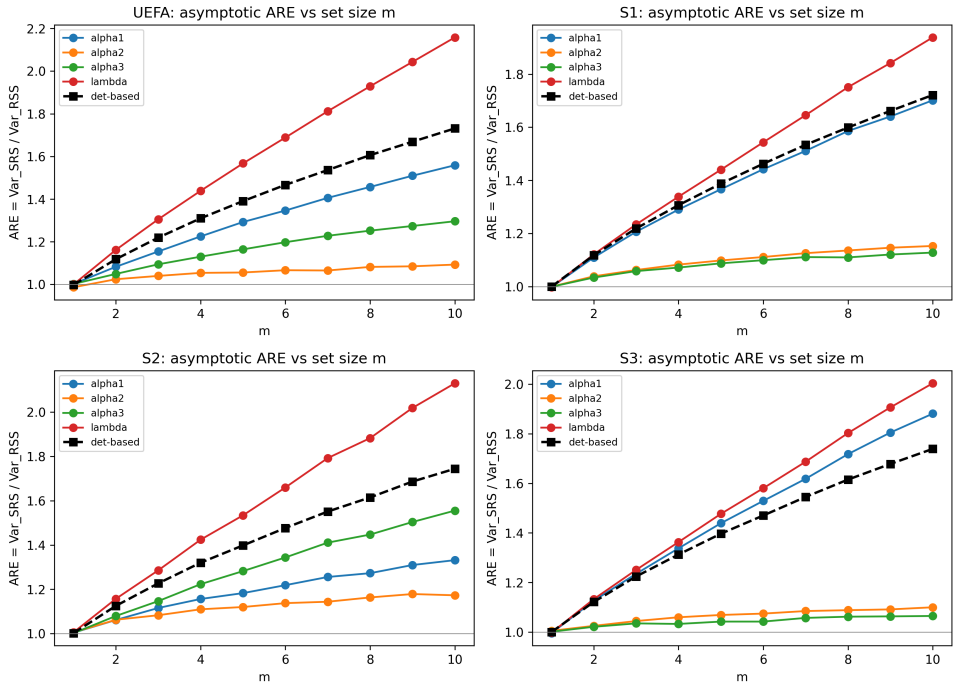


Figure 9: Asymptotic relative efficiency of RSS against SRS versus set size m for the four parameter sets.

Asymptotic normality. To check (36) under RSS, 400 RSS samples ($m = 5$, $c = 20$, $n = 100$) were generated at the UEFA parameters; Table 36 gives the summary. The empirical standard deviations exceed the asymptotic standard errors by 3–7% (ratio 1.03–1.07), a normal finite-sample effect, and the empirical correlations of the estimators match the asymptotic ones (for example $\widehat{\alpha}_1, \widehat{\lambda}$: 0.66 against 0.69; $\widehat{\alpha}_3, \widehat{\lambda}$: 0.59 against 0.54). The sampling distribution of $\widehat{\lambda}$ is close to normal, and those of $\widehat{\alpha}_1$ and $\widehat{\alpha}_3$ are right-skewed but close to log-normal (Shapiro–Wilk $p = 0.51$ and 0.61 on the log scale). The distribution of $\widehat{\alpha}_2$, the smallest parameter (0.47), is skewed on both scales. Normal-theory intervals for the shape parameters should therefore be computed on the logarithmic scale, which supports the log-Wald intervals of Section 3.4.

Table 36: Asymptotic check under RSS ($m = 5$, $n = 100$, 400 replicates): Shapiro–Wilk p-values on the original (SW) and logarithmic (SW-log) scales.

Param.	True	Mean	Emp. SD	Asym. SE	SD/SE	Skew	Ex. kurt.	SW	SW-log
α_1	1.4452	1.5005	0.2338	0.2240	1.043	0.335	−0.015	0.012	0.510
α_2	0.4681	0.4888	0.1218	0.1140	1.068	0.418	0.859	0.0005	0.000
α_3	1.1704	1.1937	0.1739	0.1619	1.075	0.302	−0.078	0.029	0.611
λ	0.0390	0.0395	0.0029	0.0028	1.027	0.205	0.121	0.438	0.962

5.4 Monte Carlo Comparison of RSS and SRS

For each parameter set, $m \in \{2, 5, 10\}$ and $n \in \{30, 50, 70, 100\}$ ($c = n/m$ cycles), $R = 1000$ RSS samples were generated and fitted by maximizing (33). The SRS results of Section 5.1 serve as the baseline, and the relative efficiencies RE_j and $\mathrm{RE}_{\mathrm{joint}}$ of (41) were computed. Table 37 compares bias and MSE for $m = 5$ and the UEFA set, Table 38 gives RE_j for $m = 2, 5, 10$ and the UEFA set, and Table 39 gives the joint relative efficiency for all four sets (Figure 10).

RSS reduces the MSE of nearly every estimator and also reduces the bias in most cells. The relative efficiency increases with the set size, from about 1.1 for $m = 2$ to 1.1–1.6 for $m = 5$ and

1.2–2.7 for $m = 10$ for the UEFA set; it is largest for λ (1.55–1.64 for $m = 5$ and 2.1–2.7 for $m = 10$) and α_1, and smallest for α_2. The joint efficiency is 1.4–1.6 for $m = 5$ and 1.7–2.1 for $m = 10$ in every set, in agreement with the asymptotic values 1.39 and 1.74 of Section 5.3. For small m some cells fall slightly below one for α_2 and α_3 (UEFA: 0.982 for α_2 and 0.977 for α_3 at $m = 2$; S3: 0.85 for α_3 at $n = 30$). Since the asymptotic AREs of these parameters are close to one at small m (Table 35) and the MSE ratio of two independent samples of 1000 replicates carries a few percent of noise, such cells indicate no detectable gain rather than a loss. For the S3 set at $m = 10$, $n = 30$ one interval length is inflated by boundary solutions (4.4% of the replicates) and should be read through the median rather than the mean.

The log-Wald coverage of the RSS intervals lies between 0.922 and 0.961 in all 48 configurations of the four sets (between 0.928 and 0.961 for UEFA), and the proportion of boundary solutions is at most 0.8% except for S3 at $n = 30$ (4.4–5.4%), as under SRS.

Table 37: SRS versus RSS ($m = 5$), UEFA parameters ($R = 1000$): bias, MSE and RE = $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{RSS}}$.

n	Param.	Bias SRS	Bias RSS	MSE SRS	MSE RSS	RE
30	α_1	0.1832	0.1290	0.3707	0.2487	1.490
	α_2	0.0626	0.0494	0.0692	0.0582	1.188
	α_3	0.1025	0.0813	0.1577	0.1148	1.373
	λ	0.00210	0.00152	5.40×10^{-5}	3.29×10^{-5}	1.642
50	α_1	0.0870	0.0719	0.1669	0.1220	1.368
	α_2	0.0194	0.0230	0.0340	0.0303	1.120
	α_3	0.0538	0.0373	0.0773	0.0601	1.287
	λ	0.00109	0.00057	2.55×10^{-5}	1.65×10^{-5}	1.551
70	α_1	0.0669	0.0652	0.1174	0.0881	1.332
	α_2	0.0175	0.0184	0.0248	0.0211	1.173
	α_3	0.0398	0.0259	0.0498	0.0410	1.217
	λ	0.00080	0.00068	1.96×10^{-5}	1.21×10^{-5}	1.627
100	α_1	0.0462	0.0348	0.0715	0.0542	1.318
	α_2	0.0137	0.0154	0.0165	0.0155	1.065
	α_3	0.0219	0.0271	0.0355	0.0302	1.175
	λ	0.00042	0.00038	1.26×10^{-5}	8.12×10^{-6}	1.551

Table 38: Relative efficiency RE_j of RSS for $m = 2, 5, 10$, UEFA parameters.

	α_1			α_2			α_3			λ		
n	2	5	10	2	5	10	2	5	10	2	5	10
30	1.126	1.490	2.132	1.074	1.188	1.373	1.103	1.373	1.638	1.252	1.642	2.718
50	1.023	1.368	1.719	0.982	1.120	1.182	1.087	1.287	1.335	1.089	1.551	2.103
70	1.212	1.332	1.761	1.157	1.173	1.371	0.977	1.217	1.344	1.198	1.627	2.180
100	1.029	1.318	1.688	1.037	1.065	1.236	1.057	1.175	1.426	1.065	1.551	2.317

Table 39: Joint relative efficiency $\mathrm{RE}_{\mathrm{joint}}$ for $m = 2, 5, 10$, all parameter sets.

	UEFA			S1			S2			S3		
n	2	5	10	2	5	10	2	5	10	2	5	10
30	1.188	1.554	2.091	1.093	1.442	1.963	1.227	1.637	2.135	1.104	1.440	1.846
50	1.112	1.470	1.807	1.224	1.492	1.881	1.044	1.383	1.808	1.146	1.392	1.828
70	1.134	1.472	1.869	1.127	1.460	1.850	1.119	1.410	1.829	1.145	1.403	1.858
100	1.107	1.385	1.806	1.118	1.351	1.702	1.094	1.380	1.774	1.111	1.377	1.749

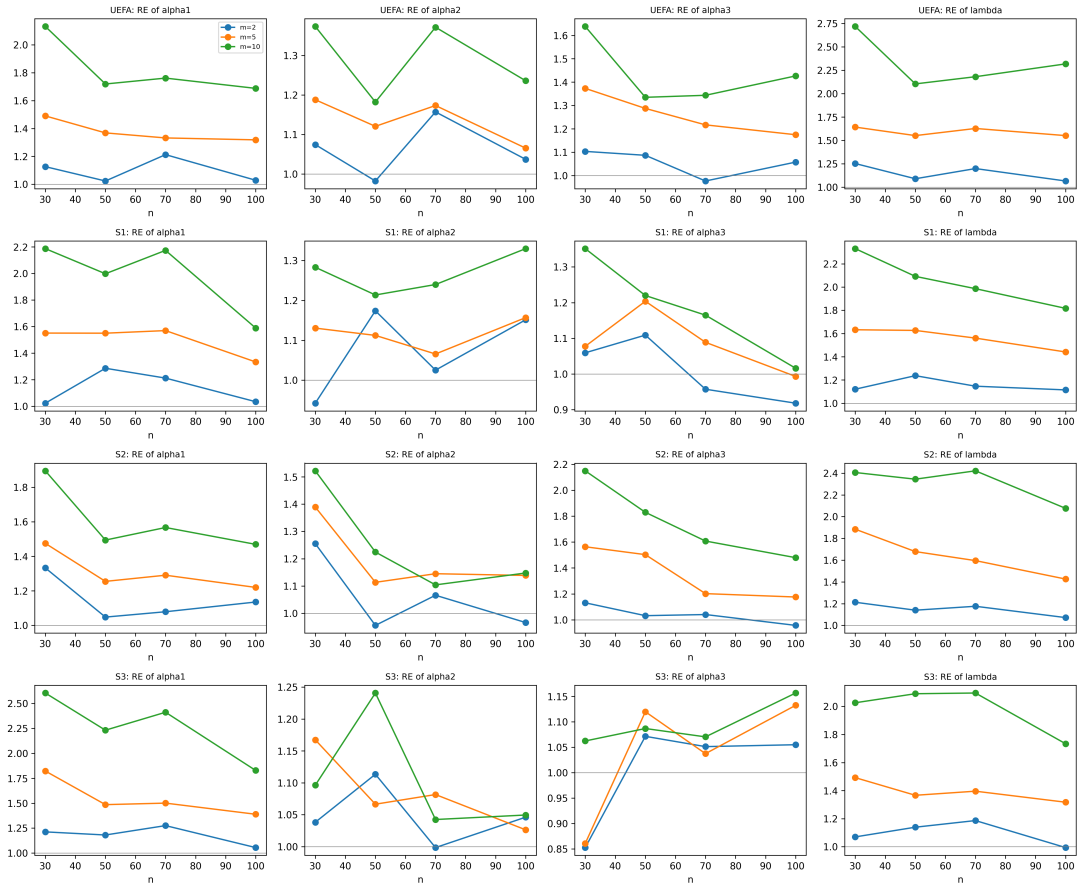


Figure 10: Relative efficiency of RSS against SRS versus n for $m = 2, 5, 10$, by parameter and parameter set.

6 Conclusion

This report studied the bivariate generalized exponential distribution of Kundu and Gupta under simple random sampling and ranked set sampling. The joint CDF and density were verified numerically, including the singular part on the diagonal, and the marginal and conditional distributions (the latter with a point mass), the survival, hazard and quantile functions, the joint and conditional moments, the covariance and correlation, and the order statistics with their concomitants were derived and tabulated. The correlation of the model is tied to the tie probability through α_3, so the model suits data in which strong dependence goes with many ties, such as the UEFA Champions League data.

Maximum likelihood estimation reproduced the published estimates on the UEFA data, with Wald, log-Wald, profile and bootstrap intervals close to the published ones, and the BVGE model was moderately preferred to the Marshall–Olkin model (AIC difference 2.33, BIC difference 0.72). A likelihood for ranked set samples ranked on the first component was derived from the joint density of an order statistic and its concomitant, and it is valid for the mixed distribution including ties.

Monte Carlo experiments showed that the estimators under SRS and RSS are accurate, with a small positive finite-sample bias that falls with n, and that the log-Wald intervals attain close to nominal coverage (0.92 to 0.96). RSS is consistently more efficient than SRS: the asymptotic determinant-based efficiency is 1.12, 1.39 and 1.74 for set sizes $m = 2, 5, 10$, the finite-sample joint efficiencies (1.4 to 1.6 for $m = 5$ and 1.7 to 2.1 for $m = 10$) agree with these values, and the gain is largest for the rate parameter λ and the shape α_1 and smallest for α_2.

Limitations and further work. The efficiency results assume perfect ranking and ranking on X_1 only, and the ranked samples are drawn from the model fitted to a small data set, because the real data cannot be used for RSS. The model has monotone marginal hazards and cannot represent strongly correlated data with few ties, and the preference over the Marshall–Olkin model on the UEFA data is only moderate. Natural extensions are imperfect ranking, ranking on the second component or on an auxiliary variable, other RSS designs (double and median ranked set sampling), censored data, and the multivariate extension of the model.

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